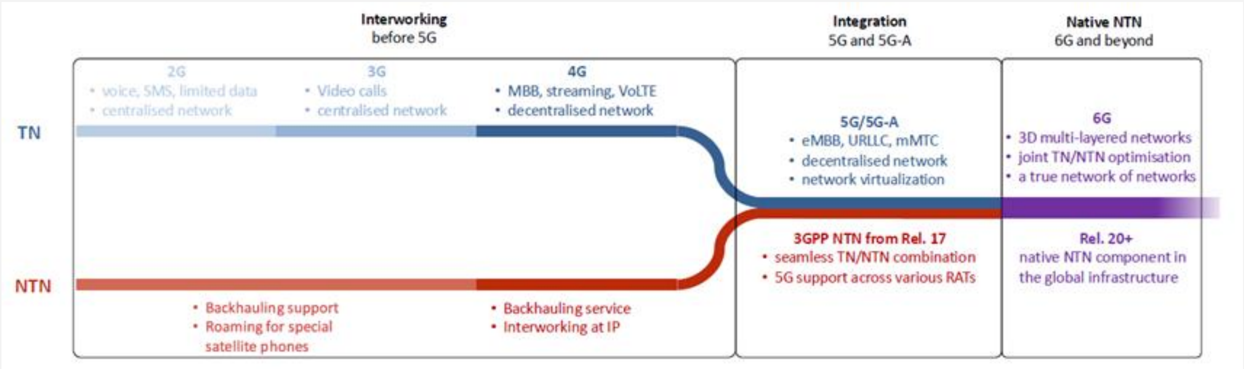
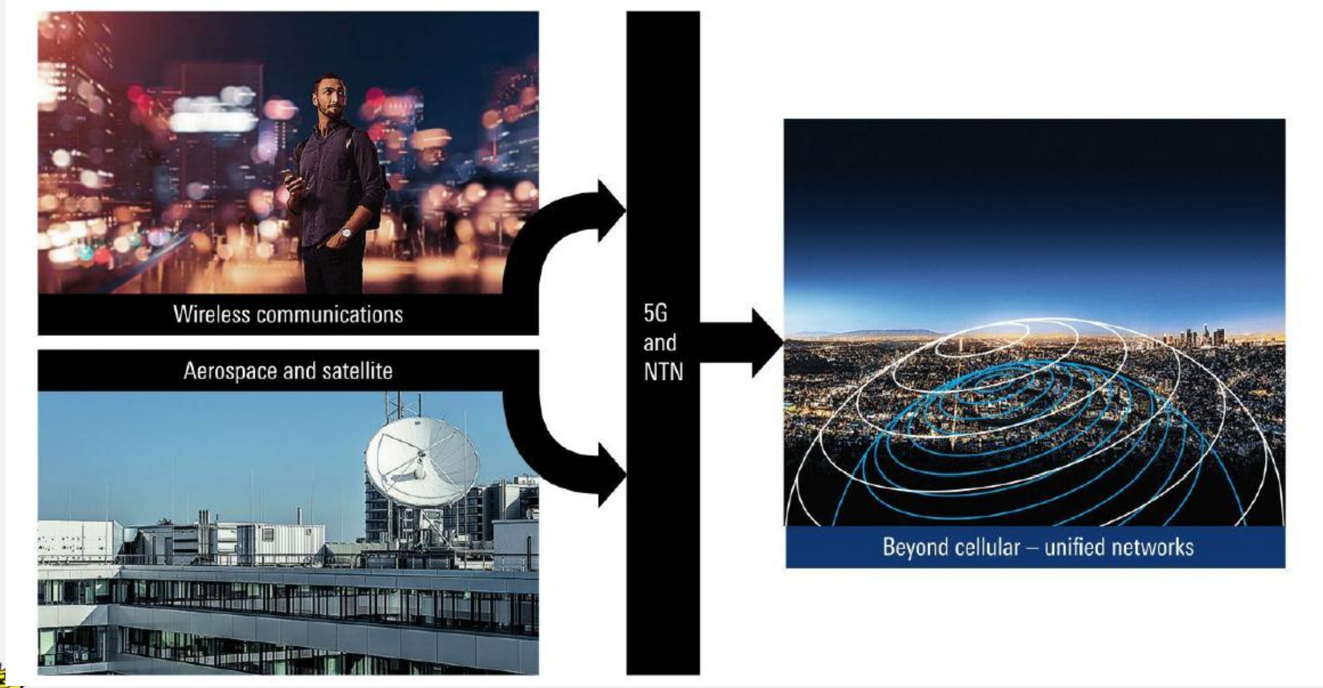
TE 456: Satellite Communication and Navigation System

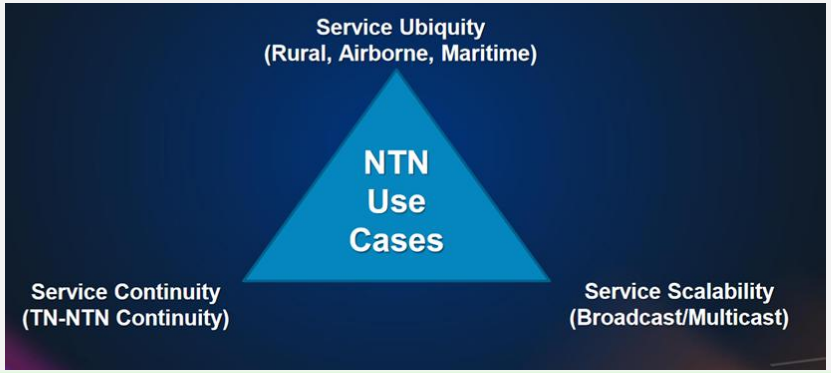
3GPP 5G NR- NTN

Introduction

- 5G and NTN

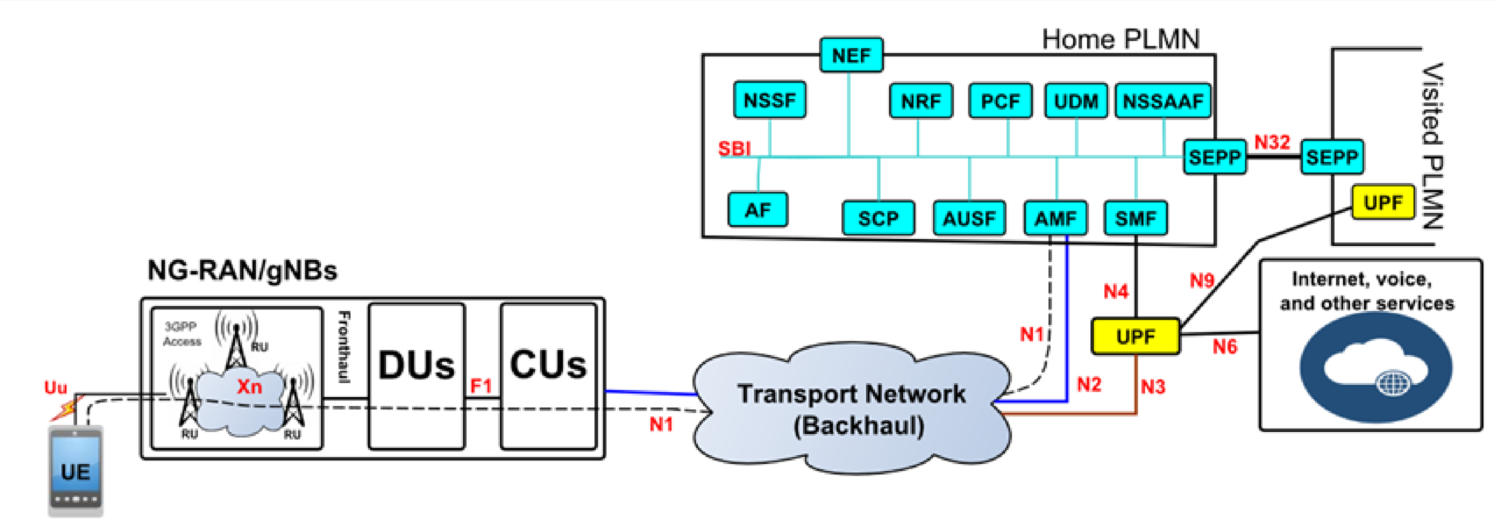






Introduction

- 5G networks



UE ↔ gNB ↔ 5GC

- are no longer just limited to cell towers on the ground

Introduction

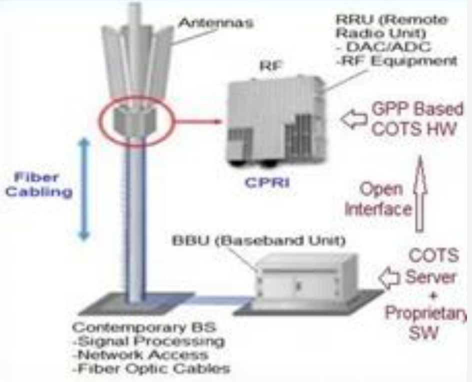
- With Non-Terrestrial Networks (NTN)
  - Satellites are now becoming a key part of 5G
    - bringing connectivity to remote areas, oceans, and even airplanes
- NTNs leverage
  - space/air base stations such as satellites, High Altitude Platforms (HAPs), and Unmanned Aerial Vehicles (UAVs)
    - for expanding wireless coverage to underserved rural/remote areas
      - supporting emergency communications, and offloading traffic in highly congested urban environments
- In this lecture
  - We will look at how NTNs are integrated into terrestrial networks
    - 3GPP 5G NR-NTN

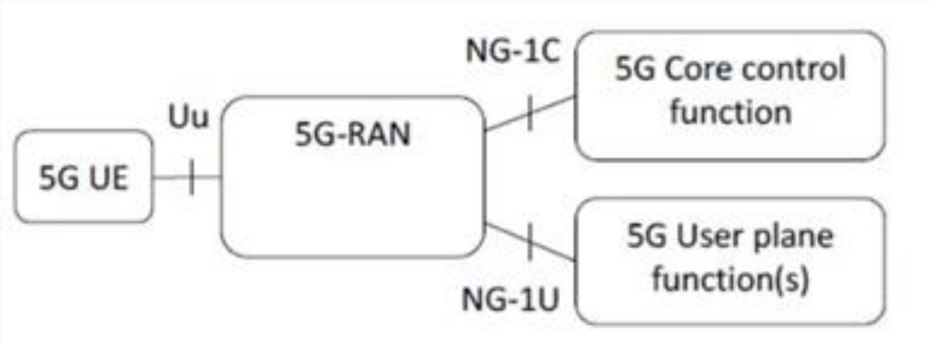
5G NR NTN – Network Elements

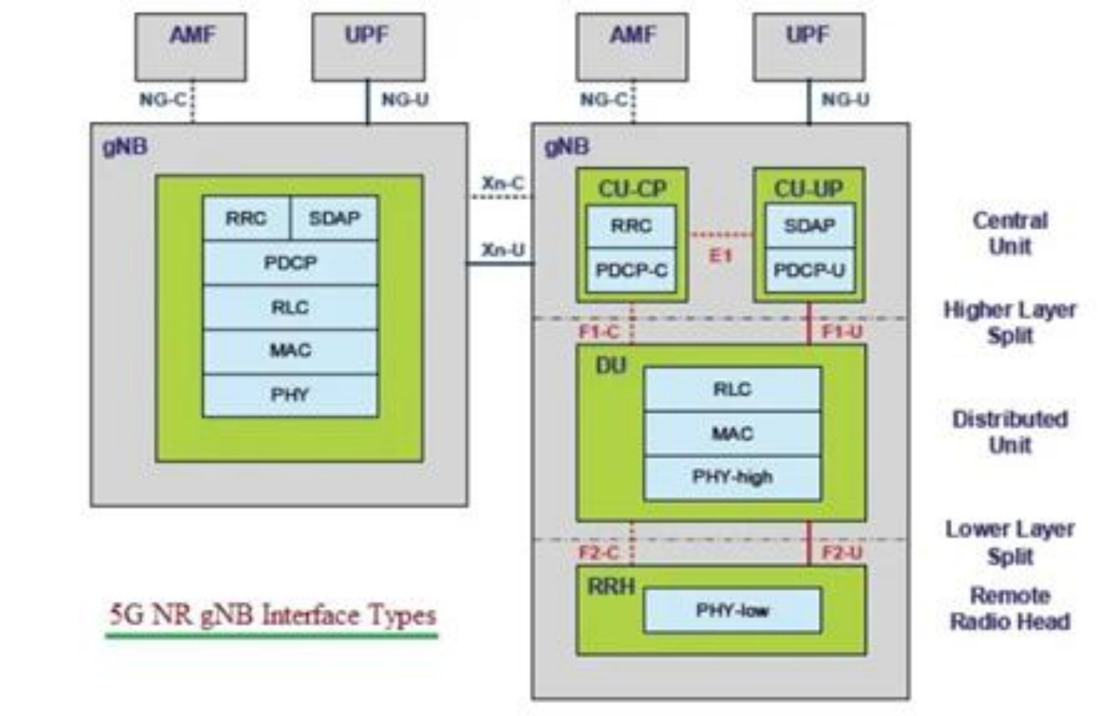
- The important elements are
  - Satellite
    - It provides NTN radio access to the UE and is connected via the feeder link (SRI) to the terrestrial 5GC functions
      - SAN (satellite access node) is the term used
  - Service link
    - is the 5G radio access provided to the UE
      - i.e. the radio interface (Uu) in a non-terrestrial network constellation
  - Feeder link or Satellite radio interface (SRI)
    - describes the connection between the satellite and the gateway
      - It operates on the same or different frequency as the service link

5G NR NTN – Network Elements

- Next Generation Radio Access Network
  a. Terrestrial Network
     - gNB (NG-RAN)
  b. Non-Terrestrial Network
     - NTN gNB (NTN NG-RAN)







5G NR gNB Interface Types

5G NR NTN – Network Elements

- gNB
  - corresponds to the 5G term gNB as a protocol anchor inheriting the protocol layers for the user and the control plane
  - includes
    a. a radio access network part to provide the radio interface (Uu) to the UE
    b. the gNB baseband functions
    c. the NG interface that connects the gNB to the 5GC functions

5G NR NTN – Network Elements

- NTN gNB
  - is a logical term introduced to separate a terrestrial RAN gNB from a gNB that provides NTN connectivity
    - The NTN gNB can be either onboard a satellite, when considering a regenerative payload,
      - or the NTN gNB may correspond to a terrestrial gNB that provides NTN connectivity via a transparent payload
  - Furthermore, the gNB can be disaggregated into smaller Functional units
    a. gNB-CU [Centralized Unit (CU)]
    b. gNB-DU [Distributed Unit (DU)]

5G NR NTN – Network Elements

- 5GC
  - is the general term for the 5G core network providing several functions like session and mobility management, user plane functions and many others
- Gateway
  - describes the terrestrial station that connects the gNB (or 5GC) functions to the satellite station
- UE and VSAT
  - Describe the terms for end user terminals

5GNTN- Frequency Spectrum aspects

- Frequency spectrum in the S-band and L-band (FR1) for NTN usage

3GPP, first NTN bands for S-band and L-band

NTN band no.	Uplink	Downlink	Duplex
n256	1980 MHz to 2010 MHz	2170 MHz to 2200 MHz	FDD
n255	1626.5 MHz to 1660.5 MHz	1525 MHz to 1559 MHz	FDD

3GPP, bandwidth and subcarrier spacing for NTN bands and RB no.

NTN band no.	SCS in kHz	5 MHz	10 MHz	15 MHz	20 MHz
256	15	yes	yes	yes	yes
	30		yes	yes	yes
	60	–	yes	yes	yes
255	15	yes	yes	yes	yes
	30		yes	yes	yes
	60		yes	yes	yes
		RB no.	RB no.	RB no.	RB no.
Max. transmission bandwidth configuration	15	25	52	79	106
	30	11	24	38	51
	60	–	11	18	24

5GNTN- Frequency Spectrum aspects

- Frequency ranges beyond 10 GHz envisaged for NTN

Band	Downlink (space to earth)	Uplink (earth to space)
Ku-band	10.7 GHz to 12.75 GHz	12.75 GHz to 13.25 GHz and 13.75 GHz to 14.5 GHz
Ka-band (GEO)	17.3 GHz to 20.2 GHz	27.0 GHz to 30.0 GHz
Ka-band (Non-GEO)	17.7 GHz to 20.2 GHz	27.0 GHz to 29.1 GHz and 29.5 GHz to 30.0 GHz
Q/V-band	37.5 GHz to 42.5 GHz, 47.5 GHz to 47.9 GHz, 48.2 GHz to 48.54 GHz, 49.44 GHz to 50.2 GHz	42.5 GHz to 43.5 GHz, 47.2 GHz to 50.2 GHz, 50.4 GHz to 52.4 GHz

5G NR-NTN-Applications

- First 5G NR-NTN specifications to support applications such as
  a. enhanced Mobile Broadband (eMBB) via 5G NR
  b. enhanced Machine-Type Communication (eMTC) via NB-IoT

5G NR-NTN-Architecture

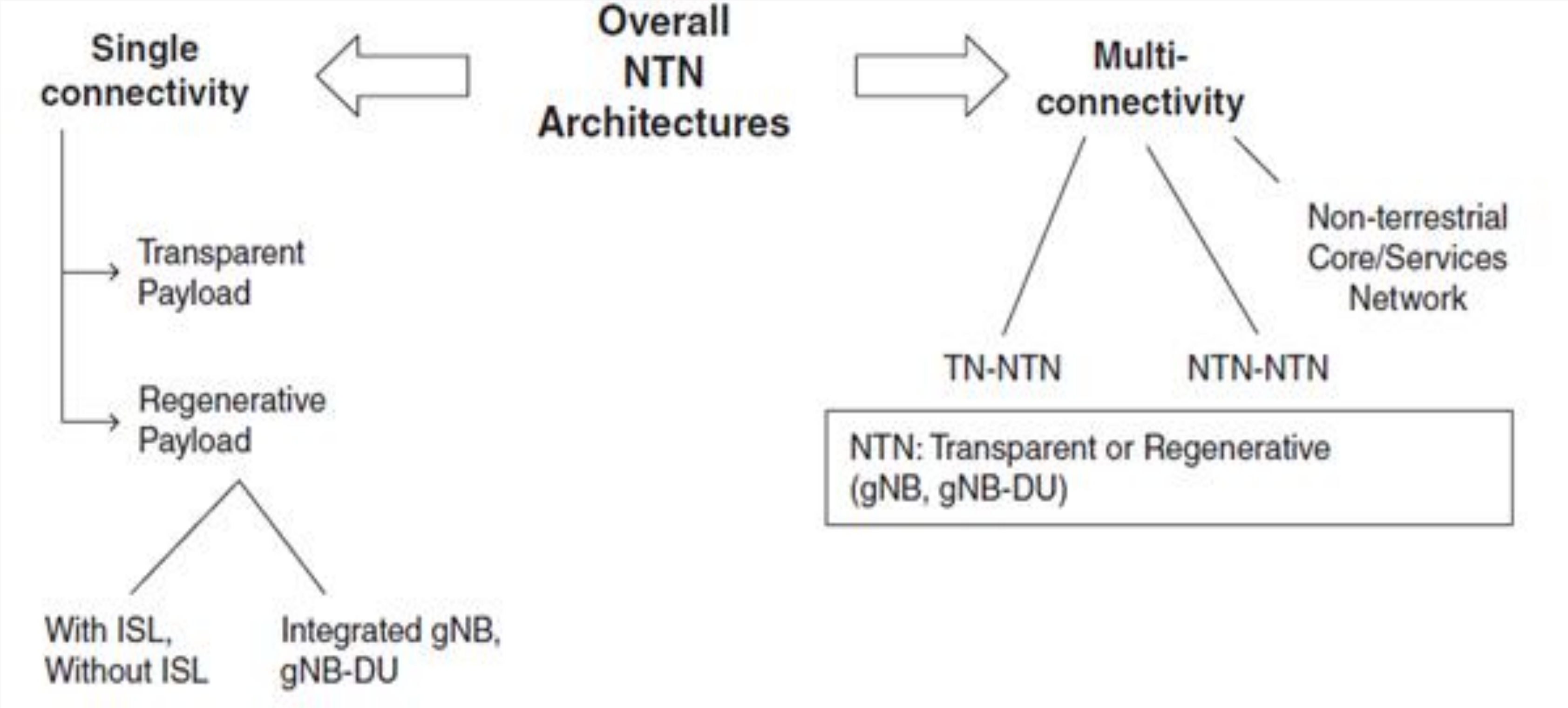
- Extension of the 5G system
  - to incorporate non-terrestrial networks (NTN)
    - entails adaptation of the existing 5G system architecture
      - Especially on the radio access network (RAN)
        - The obvious change from a terrestrial base station to an airborne or spaceborne satellite access station results in a number of amendments

5G NR-NTN – Payload Types

- The 3GPP considers two options
  1. Transparent payload (or “bent pipe payload” or “amplify-and-forward mode”)
     - In this case
       - The Next Generation NodeB (NTN gNB) is located on the ground at the gateway
     - LEO satellites relay 5G NR transmissions between
       - UEs and 5G NR base stations (NTN gNBs) located on the ground
  2. Regenerative payload (or “decode-and-forward mode”)
     - It acts as an NTN gNB from the sky

5G NTN Payload Architectures

- NTN Architecture Options



NTN Architecture Options

- A single-connectivity NTN
  - satellite or airborne communication architecture where the NTN UE connects to a single satellite or base station at a time
- In the multi-connectivity NTN
  - The NTN UE simultaneously communicates with multiple radio networks or core networks

5G NTN Payload Architectures

- The various candidate architectures include
  - Single connectivity
    - Transparent payload
    - Regenerative payload
  - Multi-connectivity NTNs
    - Transparent payload
    - Regenerative payload

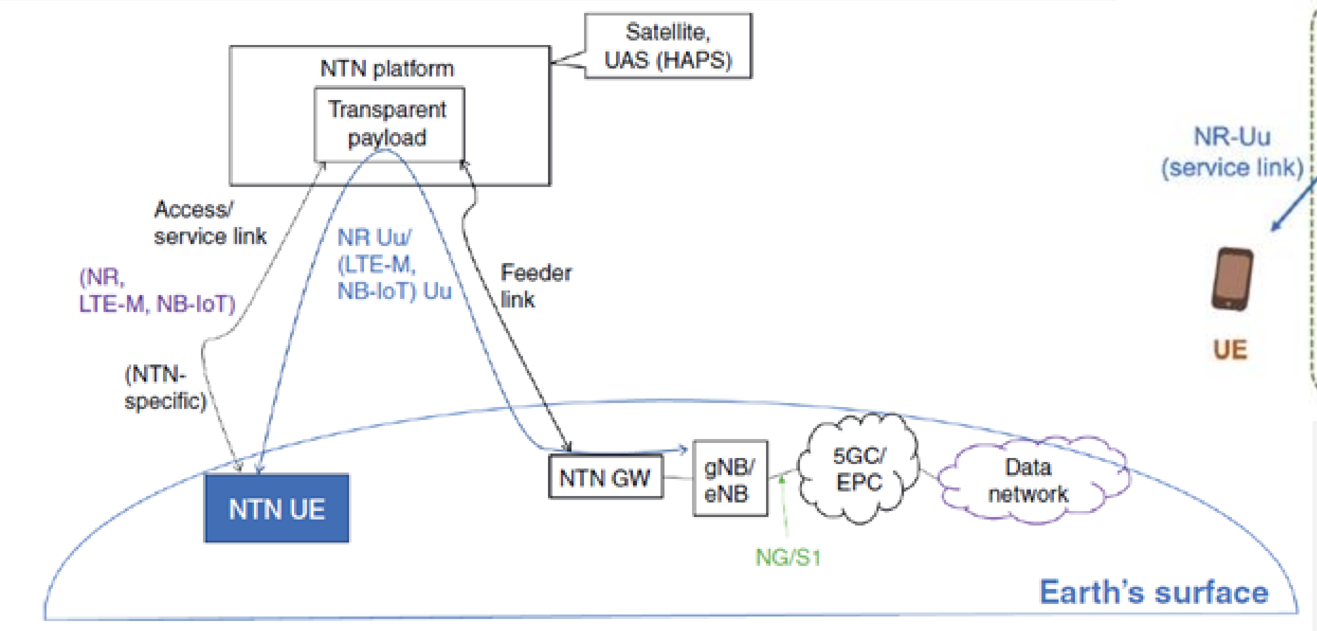
Single Connectivity Scenario

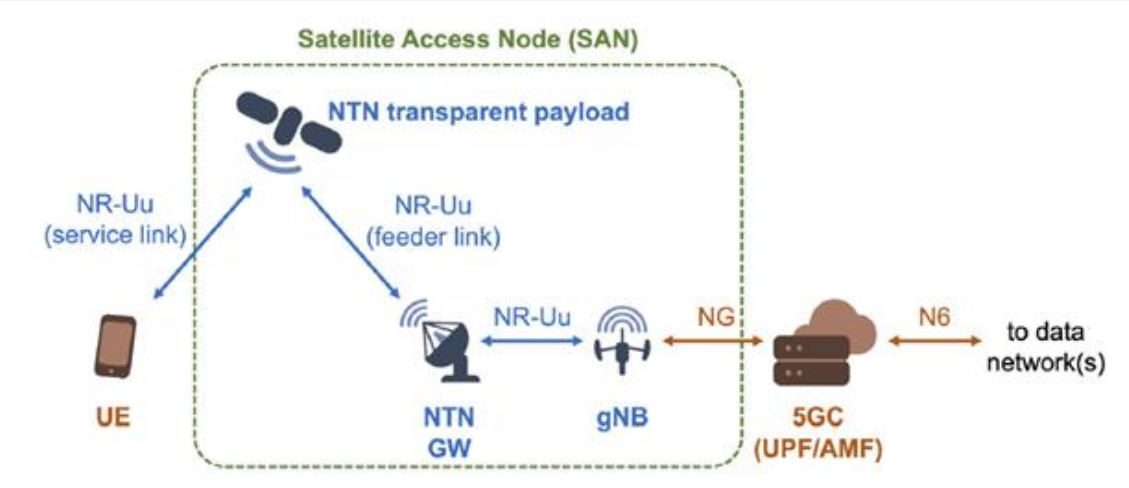
- Traditional 5G
  - UE ↔ gNB ↔ 5GC

- In the single-connectivity scenario,
  - an NTN UE may communicate with the NTN gNB (and the 5GC and potentially IP Multimedia Subsystem IMS)
    - via a transparent NTN payload or a regenerative NTN payload

5G NTN with a Transparent Payload

- In the 5G NTN architecture
  - the 5G NR-NTN UE communicates with on-the-ground gNB using a 5G NR Uu interface that passes through a transparent NTN payload and an NTN Gateway





5G NTN with a Transparent Payload

- The NR-NTN UE utilizes
  - NTN-centric enhanced 5G NR air interface to transmit and receive information from an NR-NTN gNB
- The NR-NTN UE
  - carries out suitable NR baseband processing to send information such as user traffic, or physical layer information (e.g., a random-access preamble and a scheduling request)
- The NR-NTN UE's physical layer
  - also performs suitable RF processing so the transparent NTN payload can process the 3GPP-compliant NTN-specific uplink RF signal

5G NTN with a Transparent Payload

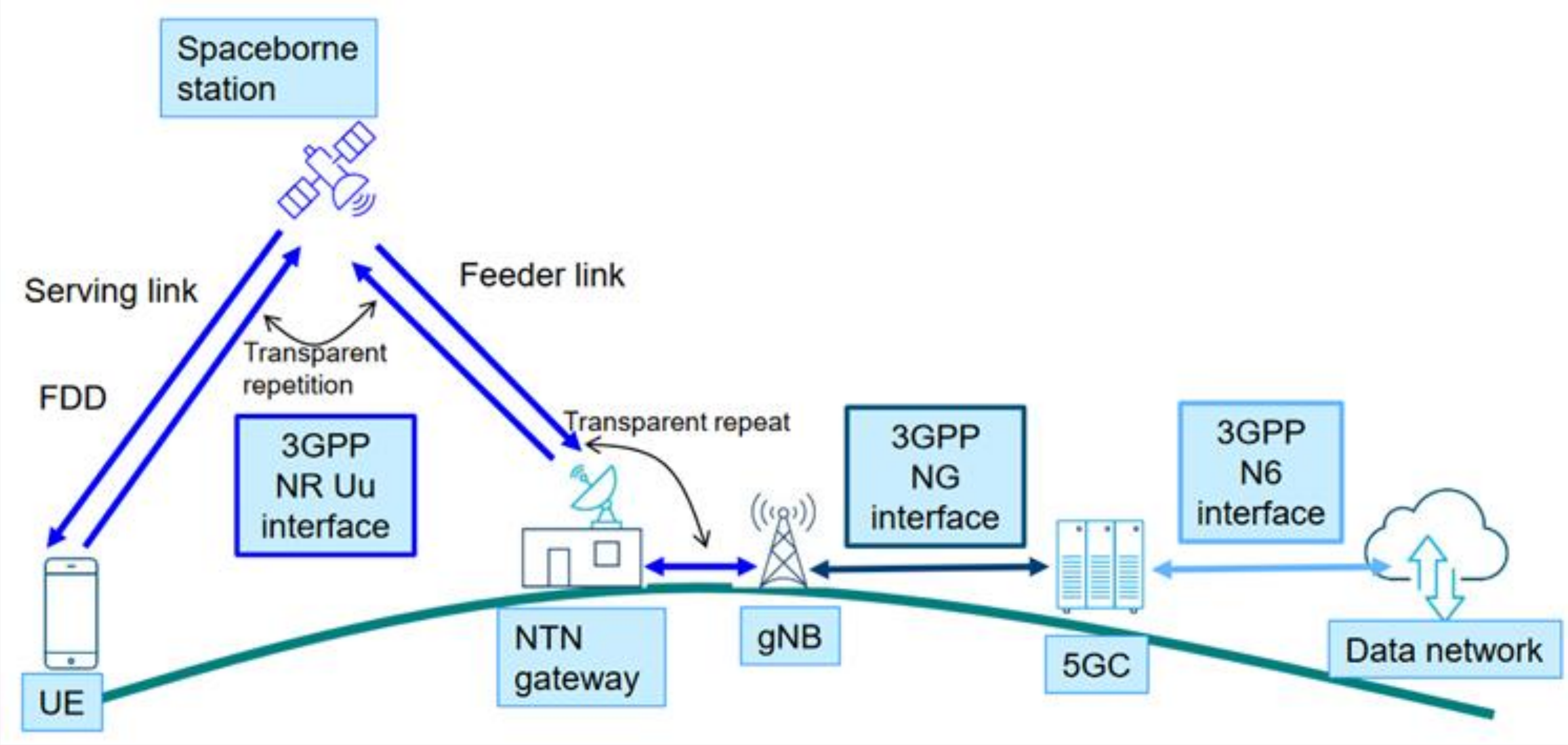
- The transparent payload
  - observes the target channel bandwidth of the uplink radio channel
    - in the NR-NTN cell to capture uplink RF signals of all NR-NTN UEs in the cell.
      - The transparent payload performs functions such as frequency translation to map the received composite NR Uu uplink signal to the carrier frequency targeted for the feeder link
  - also performs RF filtering and amplification
- The NTN Gateway
  - receives the RF signal from the transparent NTN payload via an implementation-specific feeder link
  - provides the received signal to the gNB

5G NTN with a Transparent Payload

- The NR-NTN gNB
  - carries out typical TN gNB-like baseband processing to retrieve the uplink information received from NTN Ues
  - interacts with the 5GC network functions, such as the AMF and the UPF, as needed per NTN UE

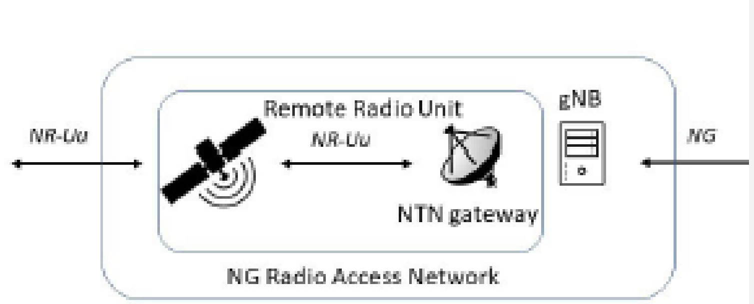
5G NTN with a Transparent Payload

- Transparent



5G NTN with a Transparent Payload

- The legacy term “base station” is disaggregated into
  i. Satellite
  ii. ground gateway
  iii. gNB functions



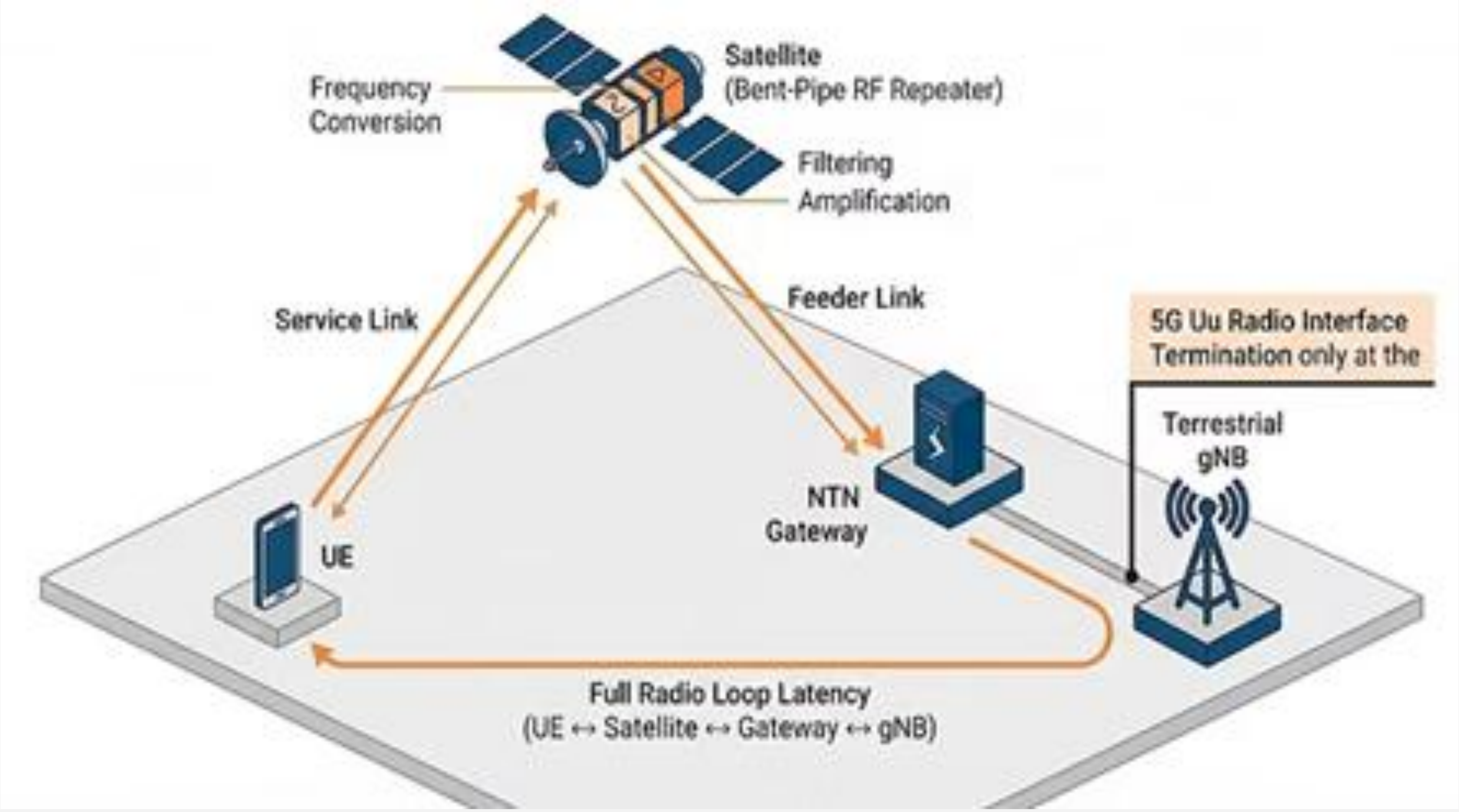
- Satellite
  - The satellite functions implement
    - RF filtering, frequency conversion, RF amplification and RF transmission and reception in both the uplink and downlink direction

5G NTN with a Transparent Payload

- **Signal Flow:** The 5G NR signal from the User Equipment (UE) traverses the **Service Link** to the satellite
  - The satellite performs frequency conversion (uplink to downlink frequencies), filtering, and amplification, then relays the signal via the **Feeder Link** to a terrestrial **NTN Gateway**, which connects to the gNB
- **Protocol Termination:** Crucially, the 5G Uu radio interface terminates at the **terrestrial gNB**, not at the satellite
  - Consequently, the radio loop latency includes the entire path:
  - UE ↔ Satellite ↔ Gateway ↔ gNB

5G NTN with a Transparent Payload

- Transparent



5G NTN with a Transparent Payload

- The functionality of the transparent payload is defined
  - as the electromagnetic waves that are transmitted from the earth's surface
    - and converted by a satellite receive antenna into an electric signal, which is channel-filtered and amplified by a low-noise amplifier (LNA).
      - The signal may be frequency-converted and a high-power amplifier (HPA) finally delivers
        - the signal to a transmitting antenna generating a reconditioned electromagnetic wave towards the earth's surface where the receive gateway is located

5G NTN with a Transparent Payload

- The termination of the 5G NR air interface (Uu)
  - with respect to the protocol anchor is in the terrestrial gNB function
    - This represents an important topic with respect to latency or RTT aspects as the one-way latency includes the serving and feeder link
      - The gateway function has no explicit task with respect to 5G NR
- The connection between
  - the UE and the terrestrial gNB includes at least the serving and the feeder link
    - One specific UE data connection to the 5GC is routed via a single gateway or gNB, exceptions are the multi connectivity scenarios

5G NTN with a Transparent Payload

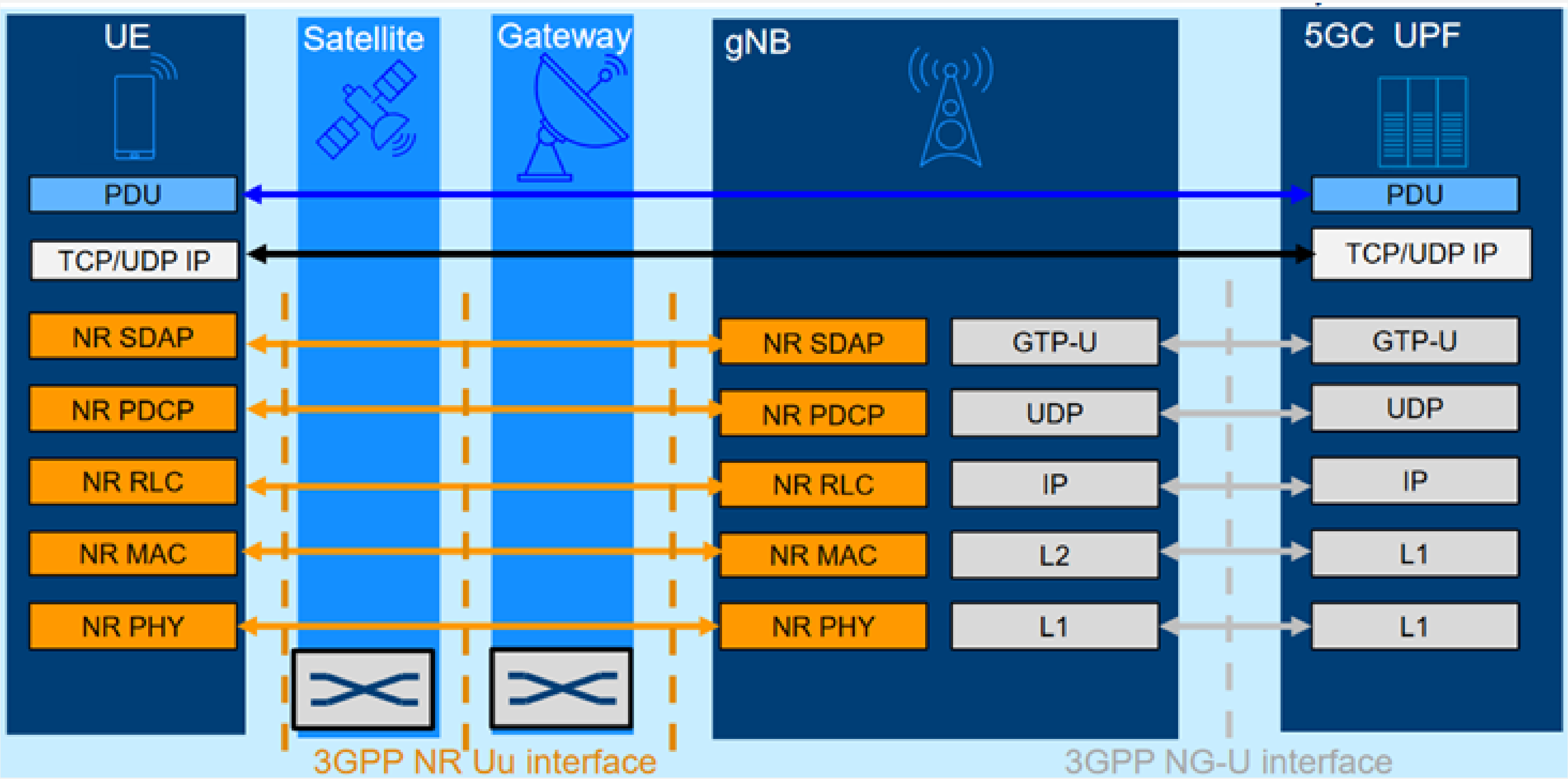
- Transparent payload:
  - the satellite merely acts as an amplify-and-forward relay between the feeder link and the service link, with no onboard processing capabilities (transparent or bent-pipe payload)
    - In this case, the gNB is deployed on the ground, typically co-located with the gateway
  - It functions as a simple relay between the UE and the ground-gNBs
    - performing radio frequency filtering, frequency conversion, and amplification without altering the waveform

5G NTN with a Transparent Payload

- With respect to the 5G NR protocol architecture
  - there is not much difference between the terrestrial RAT layer model and NTN
    - The only differences are the additional intermediate entities, satellite and ground gateway in the protocol layer model
      - which are transparent to the data flow

5G NTN with a Transparent Payload

- Protocol stack - UP

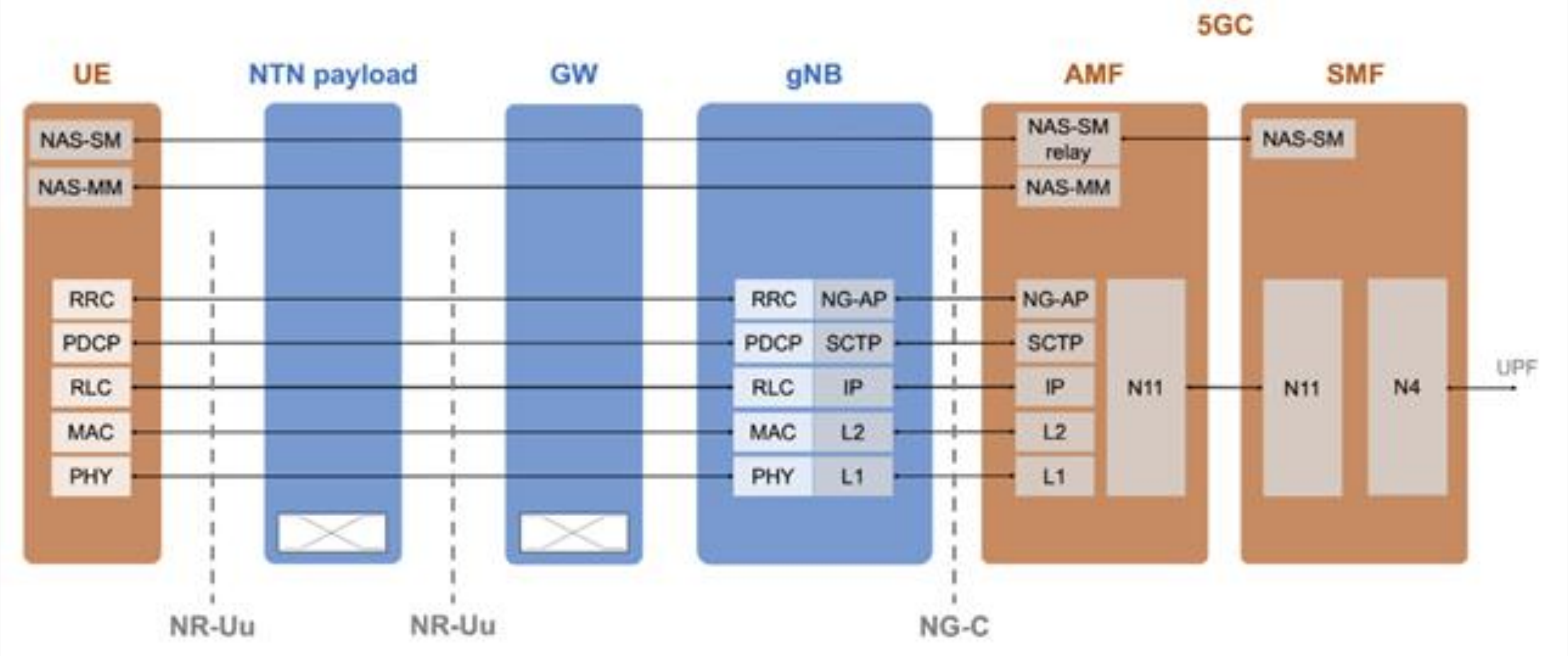


5G NTN with a Transparent Payload

- Both the satellite payload and the NTN gateway
  - are transparent with respect to the CP protocols
    - so the termination points for Radio Resource Control (RRC), Packet Data Convergence Protocol (PDCP), Radio Link Control (RLC), Medium Access Control (MAC), and Physical layer (PHY) are on the
      - NTN UE and the NTN gNB

5G NTN with a Transparent Payload

- CP protocol stack of the NTN architecture with transparent payload and direct access



5G NTN with a Transparent Payload

- Both the satellite payload and the NTN gateway
  - are transparent with respect to the UP protocols,
    - so the termination points for Service Data Adaptation Protocol (SDAP), PDCP, RLC, MAC, and PHY are on the
      - NTN UE and the NTN gNB

5G NTN with a Regenerative Payload

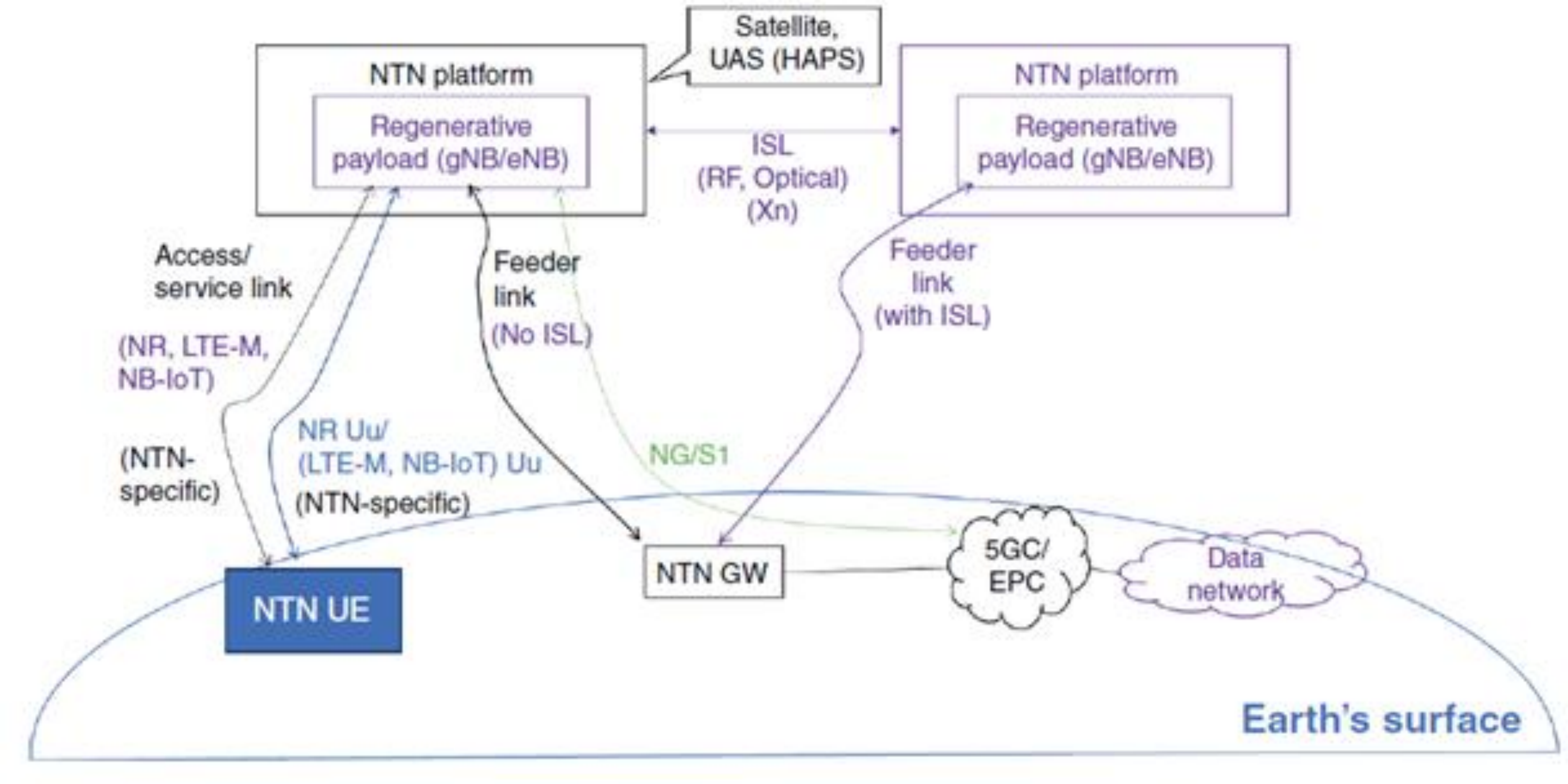
- The regenerative architecture deals
  - with the three components satellite, gateway and gNB
    - The major difference to the transparent payload architecture is that the gNB functions are incorporated into the satellite itself
      - The regenerative architecture model raises the complexity of the satellite hardware and computing power
        - Further separations are made in the optional disaggregation of the gNB functions into the distributed unit (DU) and the centralized unit (CU)

5G NTN with a Regenerative Payload

- The regenerative payload architecture
  - supports both architecture models:
    a. The satellite node can directly incorporate the DU and CU, and via the feeder link to the gateway there is a direct connection to the 5G core network (5GC)
    b. Alternatively, the satellite can only host the DU functions of the gNB and therefore the feeder link to the terrestrial gateway carries the F1 interface over the SRI and the CU is terrestrial based
- When the
  - regenerative NTN payload houses the gNB functions,
    - two NTN payloads may have an inter-satellite link (ISL) between them.
      - An ISL enables a series of NTN payloads to connect to each other in a chain to allow connectivity with a distant NTN Gateway on the ground

5G NTN with a Regenerative Payload

- In this architecture
  - The regenerative payload houses the NTN gNB



5G NTN with a Regenerative Payload

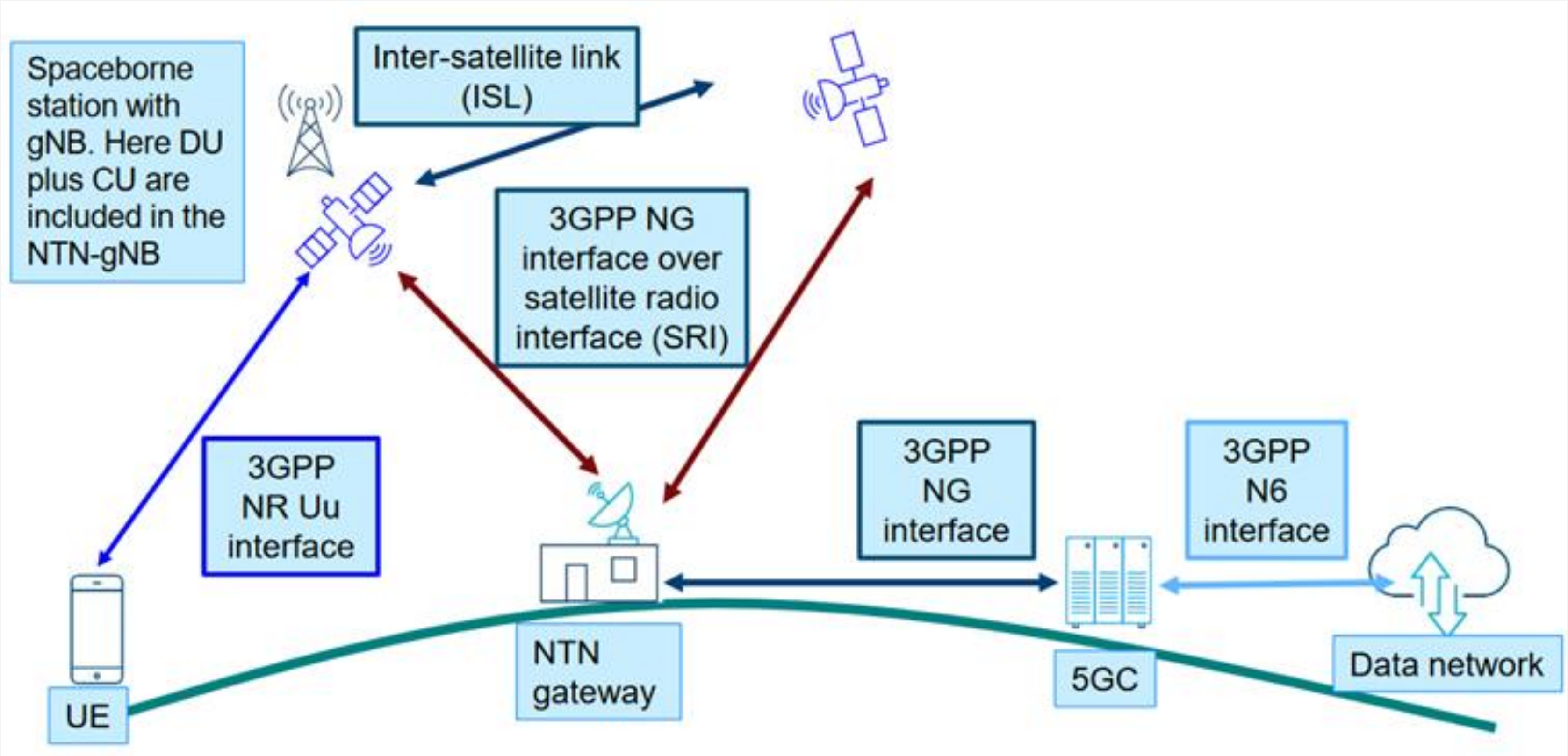
- The 5G NR-NTN UE
  - communicates with the NR-NTN gNB residing on the regenerative NTN payload using 5G NR Uu interface on the service link
- The feeder link carries
  - the NG interface information
- The NTN Gateway interfaces with

  (i) the AMF via the N2 interface and

  (ii) the UPF via the N3 interface

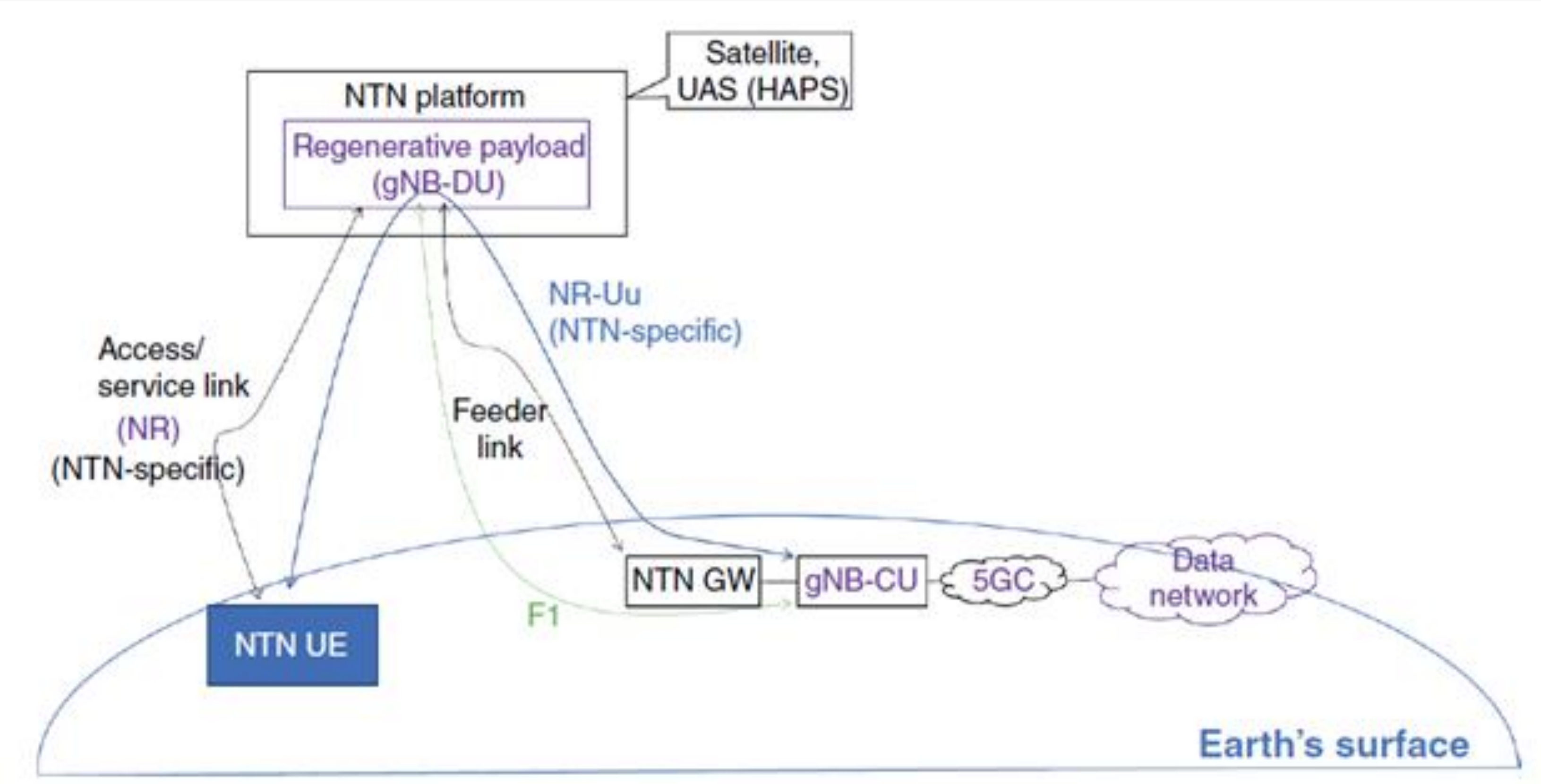
5G NTN with a Regenerative Payload

- Full gNB on satellite



5G NTN with a Regenerative Payload

- An NTN with a Regenerative NTN Payload Containing a gNB-DU

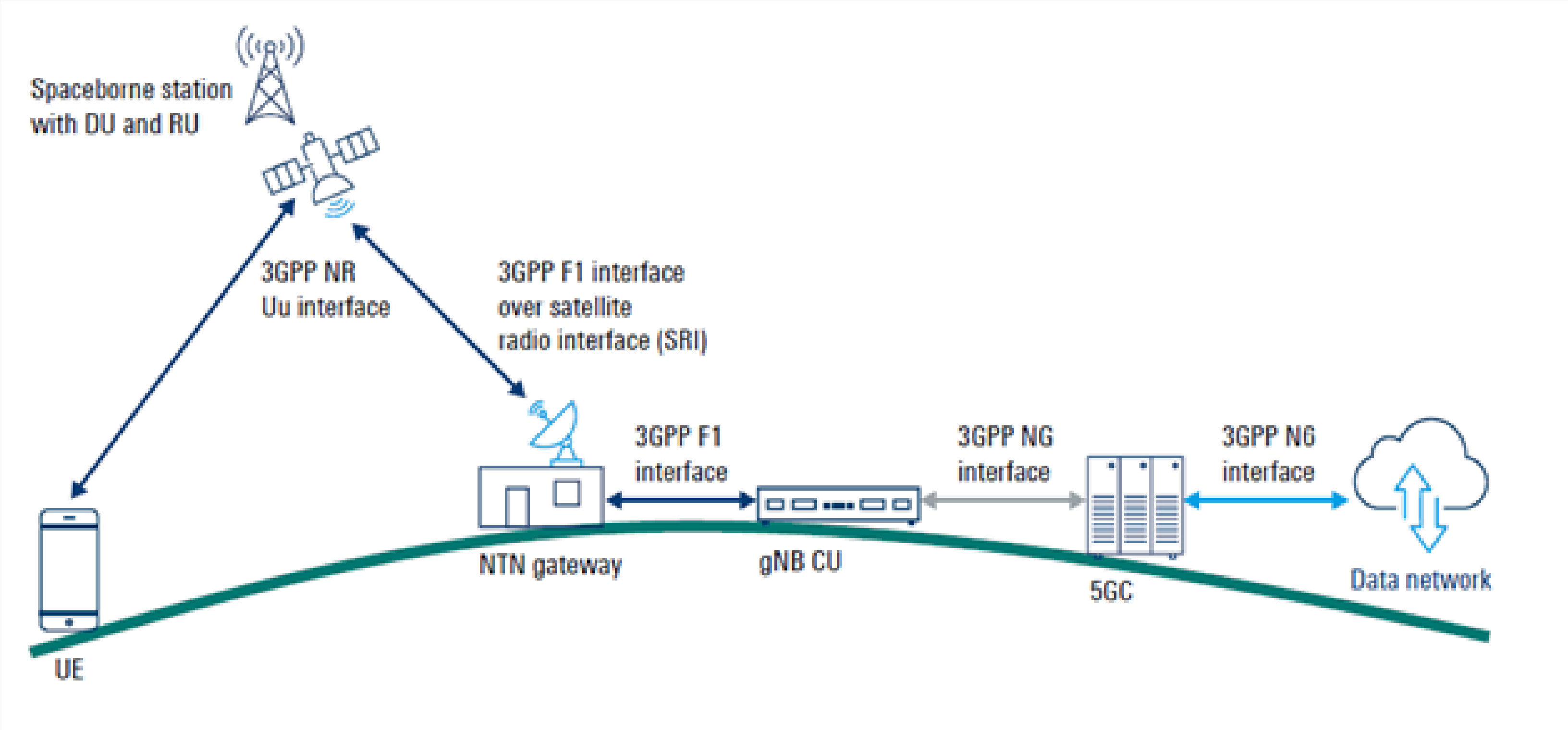


5G NTN with a Regenerative Payload

- For NR-NTN
  - with a regenerative payload housing the gNB-DU
    - The NTN gNB-CU is on the ground
- The 5G NR-NTN UE
  - communicates with the NR-NTN gNB-DU residing on the regenerative NTN payload using the 5G NR Uu interface on the service link or the access link
- The feeder link
  - carries the F1 interface information, such as F1-C signaling and F1-U user plane traffic between the NTN Gateway and the gNB-DU
- The NTN gNB-CU
  - can communicate with the core network using an on-the-ground transport network

5G NTN with a Regenerative Payload

- DU on-board and CU on ground



5G NTN with a Regenerative Payload

- The regenerative payload functionality
  - is the implementation, where an on-board processor (OBP) is inserted between the LNA and the HPA
    - This OBP allows conversion of the air interface between the uplink and the downlink
      - It allows bits or packet errors to be corrected before they are retransmitted, or packets to be routed between beams
        - Ultimately any network function can be implemented, at the expense of power and mass
          - thanks to an OBP (including gNB CUs or DUs, or any function attached to a 5GC)

5G NTN with a Regenerative Payload

- One advantage of the regenerative architecture
  - is the reduction of latency as the Uu interface terminates in the satellite that incorporates the entire NTN gNB functionality
    - Thus, the single-way latency only includes the service link
- A second advantage is the possible deployment
  - of an NTN with multiple ISL connections between the satellites, especially a swarm LEO constellation where only a few LEO satellites have access to the gateway

5G NTN with a Regenerative Payload

- advanced satellite stations
  - with on-board processing capabilities, including signal regeneration, (de)modulation, decoding, and switching/routing.
    - In this sense, the satellite effectively operates as a decode-and-forward gNB (regenerative payload)
- **Full gNB on Satellite:**
  - The satellite contains both Centralized Unit (CU) and Distributed Unit (DU), connecting directly to the 5G Core (5GC) on the ground.
- **gNB-DU on Satellite:**
  - The satellite hosts the DU (PHY/MAC/RLC layers), while the CU remains terrestrial

5G NTN with a Regenerative Payload

- **On-board processing:**
  - The satellite houses full or partial gNB functions
    - It terminates the 5G radio interface (NR-Uu) in orbit
- Options
  a. **Complete gNB on Satellite:** All gNB functions (both Central Unit and Distributed Unit) are located entirely onboard the satellite.
  b. **Split gNB Architecture:** The gNB-Central Unit (CU) remains on the ground, while the gNB-Distributed Unit (DU) is flown on the satellite, connected via standardized interfaces

5G NTN with a Regenerative Payload

- **Lower Layers (DU):**
  - The Distributed Unit (DU), which handles real-time Radio Link Control (RLC), Medium Access Control (MAC), and Physical (PHY) layers, is usually housed directly on the satellite
- **Higher Layers (CU):**
  - The Centralized Unit (CU), which handles computation-heavy Radio Resource Control (RRC) and Packet Data Convergence Protocol (PDCP), is normally kept at an Earth station or Ground Gateway

5G NTN with a Regenerative Payload

- **Technical Advantages:**
  - **Latency Reduction:**
    - By terminating the Uu interface at the satellite, latency-sensitive loops like HARQ are handled in orbit, drastically reducing RTT
  - **Inter-Satellite Links (ISL):**
    - Regenerative payloads enable mesh networking in space, allowing data routing between satellites without bouncing back to earth at every hop
    - Onboard routing enables satellites to communicate directly via optical or RF links
      - forwarding packetized data to other in-orbit gNBs before transmitting it to a terrestrial NTN Gateway

5G NTN with a Regenerative Payload

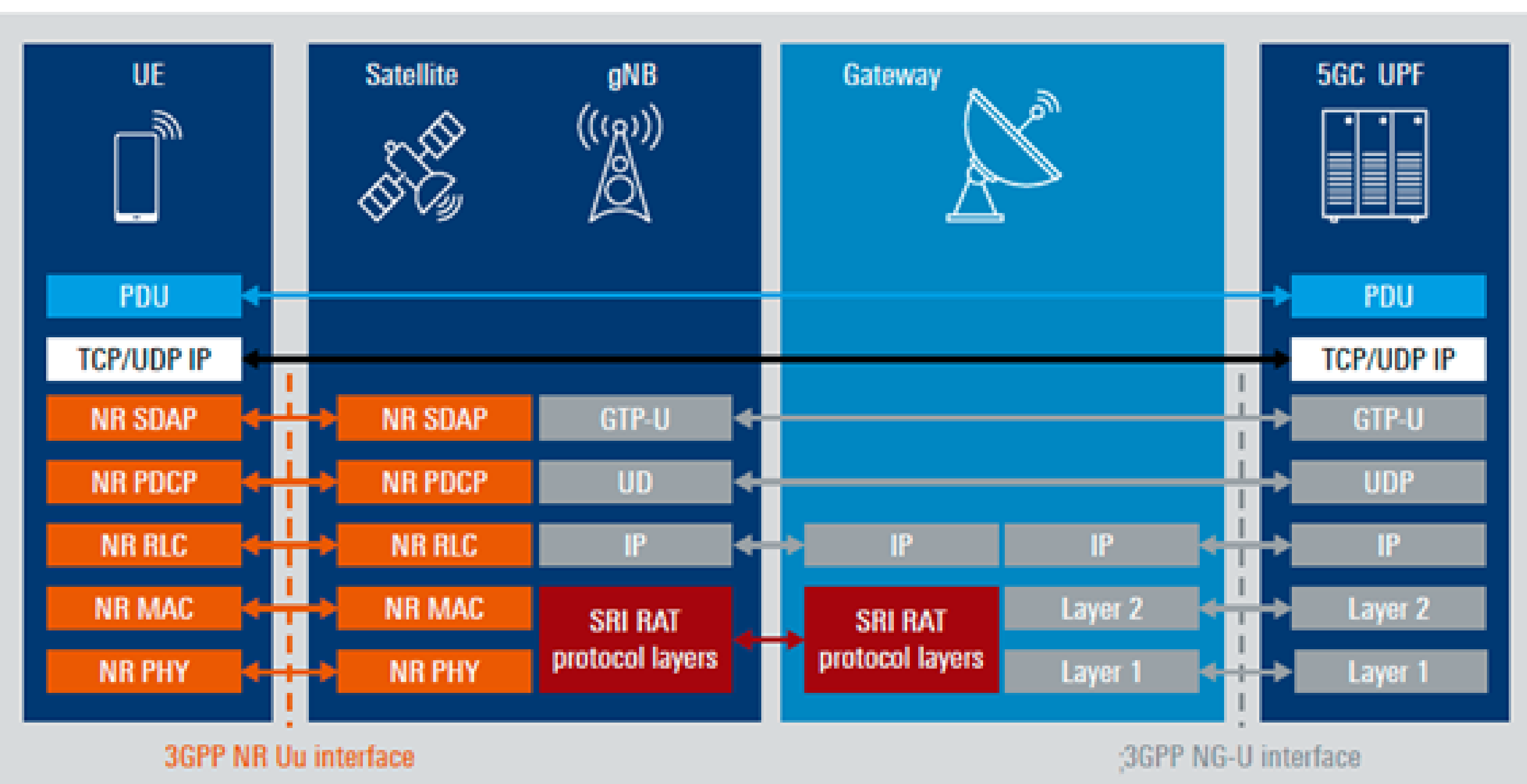
- In the protocol layer architecture
  - The Uu interface terminates in the satellite containing all the NTN gNB functions and that there is the SRI link to
    - transport the user and control plane data from the regenerative mode satellite via the gateway to the 5GC
      - To enable transparent user plane data transport
        - the user PDUs are transported over the GTP-U protocol
          - in a tunnel procedure between the satellite containing all the gNB functions and the 5GC

5G NTN with a Regenerative Payload

- The control plane architecture is similar to the user plane
  - The RRC terminates within the satellite gNB function, so here too we assume a shorter reaction time for RRC-specific radio link control procedures
    - In the case of the alternative disaggregated architecture where the satellite carries the DU functions and the CU is still on ground
      - the control plane like the RRC protocol layer also terminates on ground, hence the long delay like that in the transparent architecture prevails

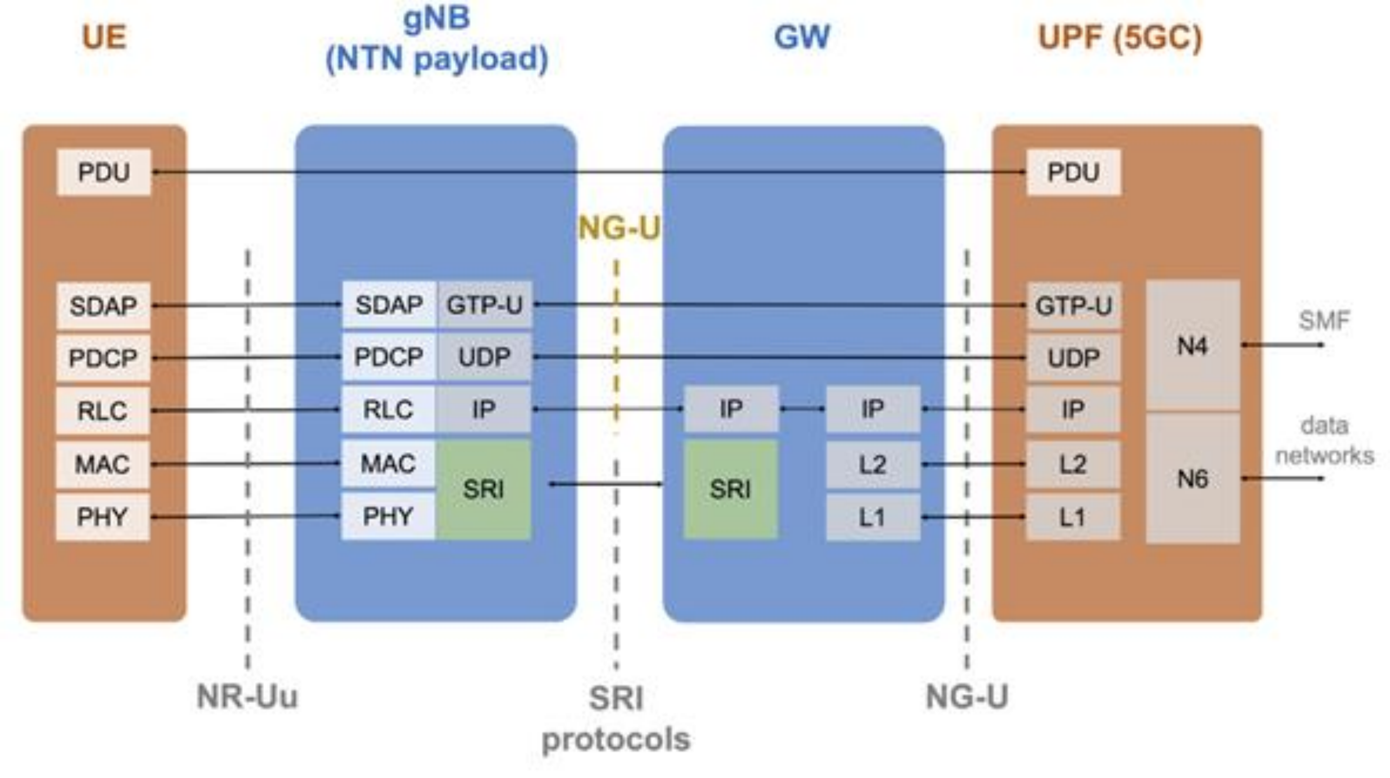
5G NTN with a Regenerative Payload

- User plane protocol



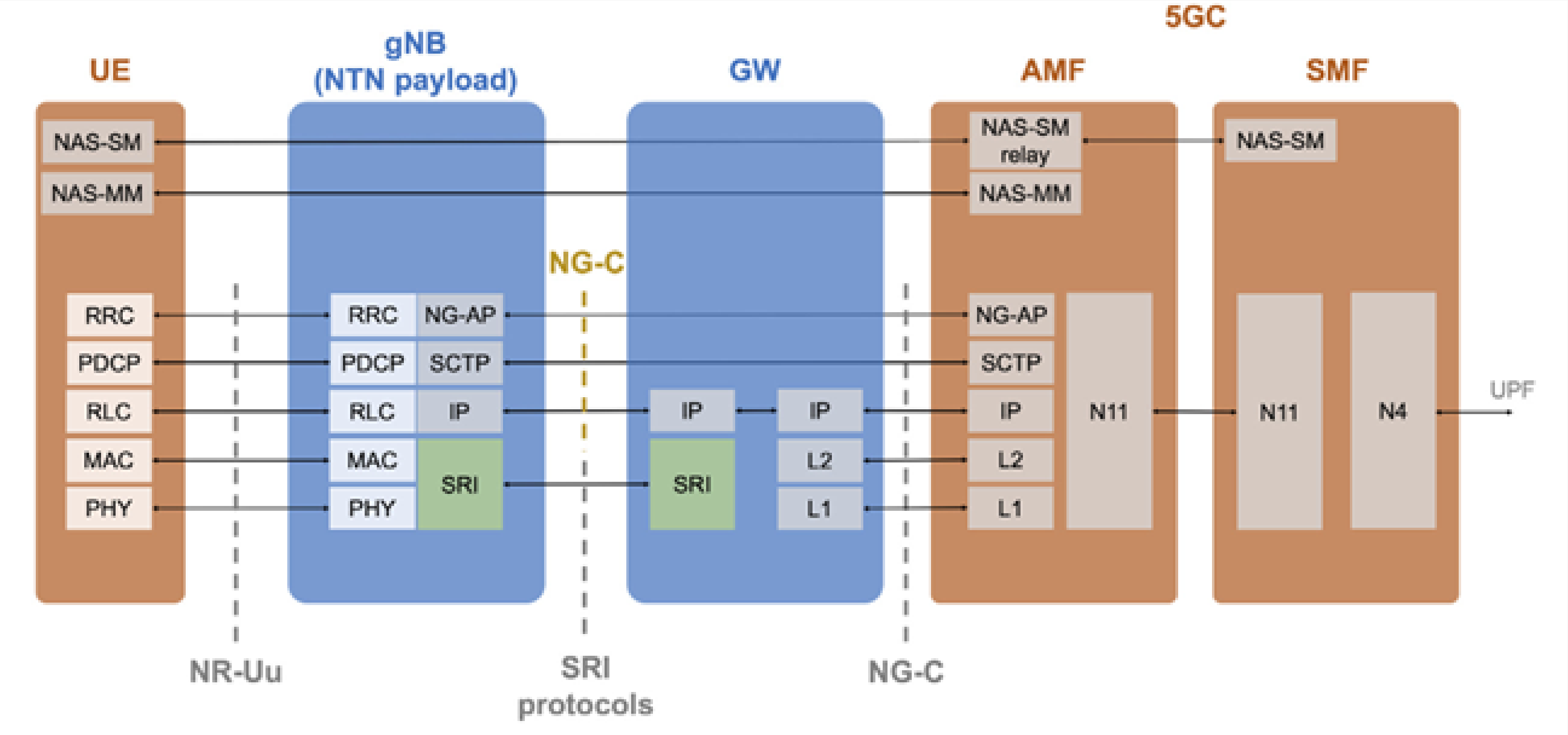
5G NTN with a Regenerative Payload

- UP protocol stack of the NTN architecture with regenerative payload (full gNB) and direct access



5G NTN with a Regenerative Payload

- CP protocol stack of the NTN architecture with regenerative payload (full gNB) and direct access

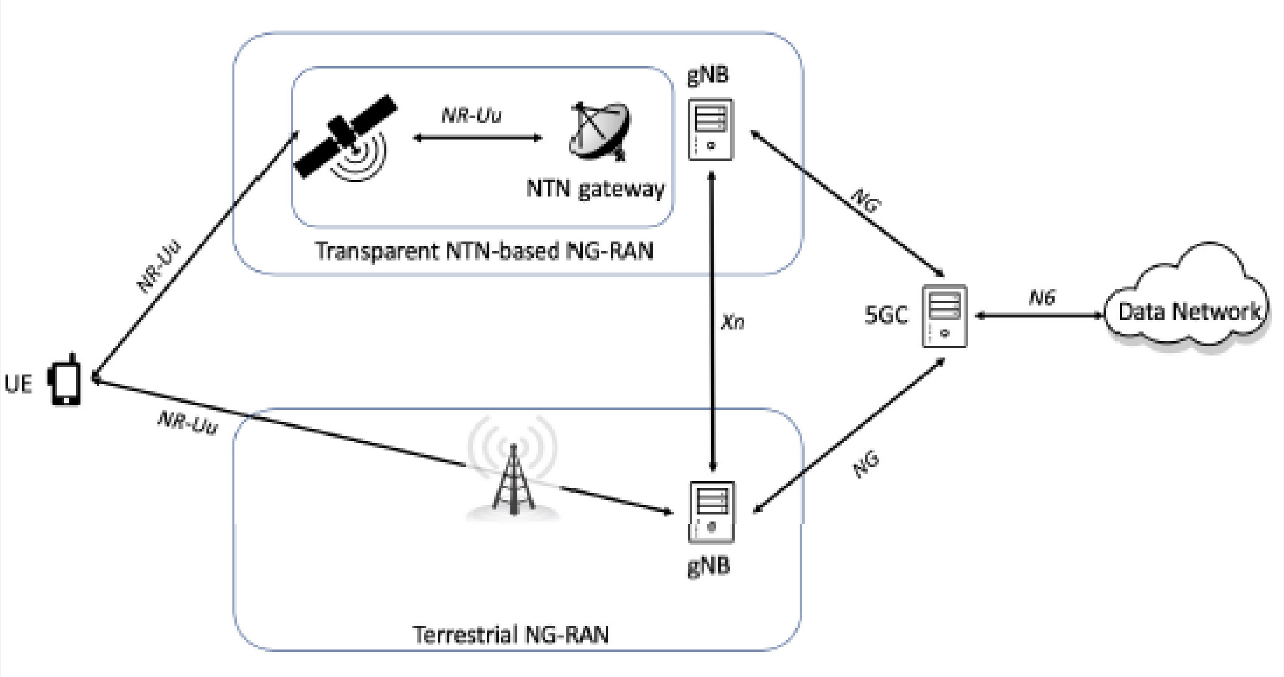


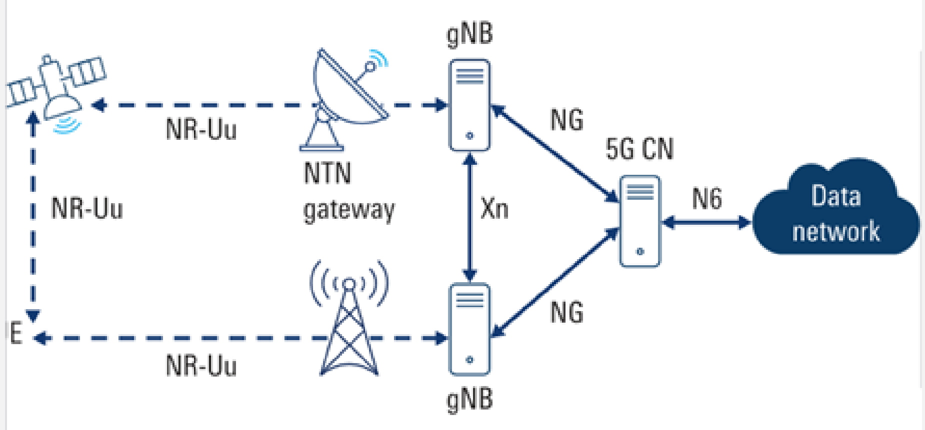
Service Continuity & Multi-Connectivity

- The integration of NTNs and terrestrial networks
  - is essential to guarantee service continuity and scalability in 5G and beyond systems
    - Service continuity can be between
      a. NG-RAN and NTN NG-RAN
      b. Two NTN NG-RAN
- Multi-connectivity
  - Allows simultaneous access to both the NTN and TN NG-RANs
    - Or two NTN NG-RANs

Service Continuity & Multi-Connectivity

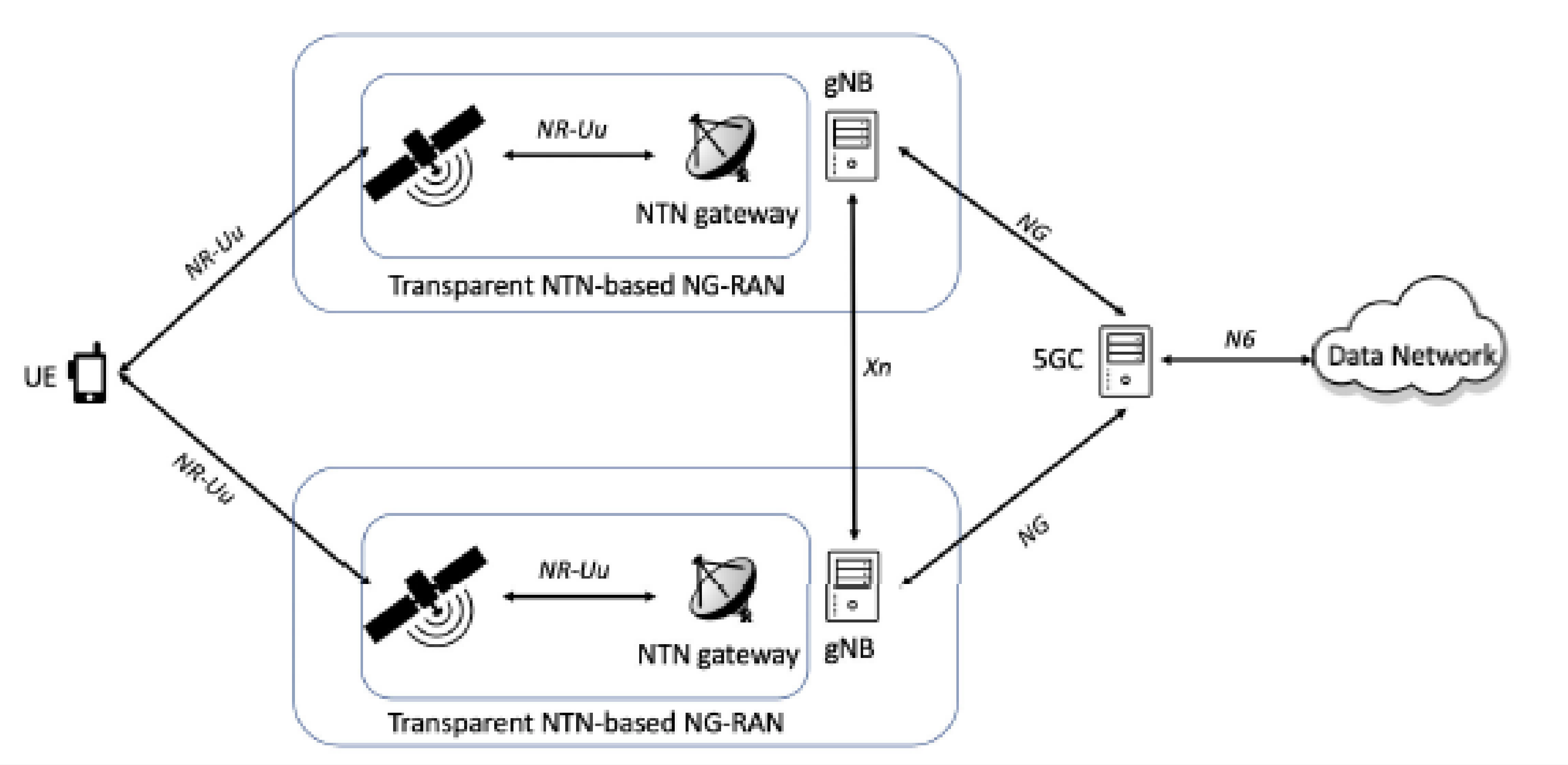
- The User terminal is connected simultaneously
  - to the 5GC via both transparent NTN-based NG-RAN and terrestrial NG-RAN





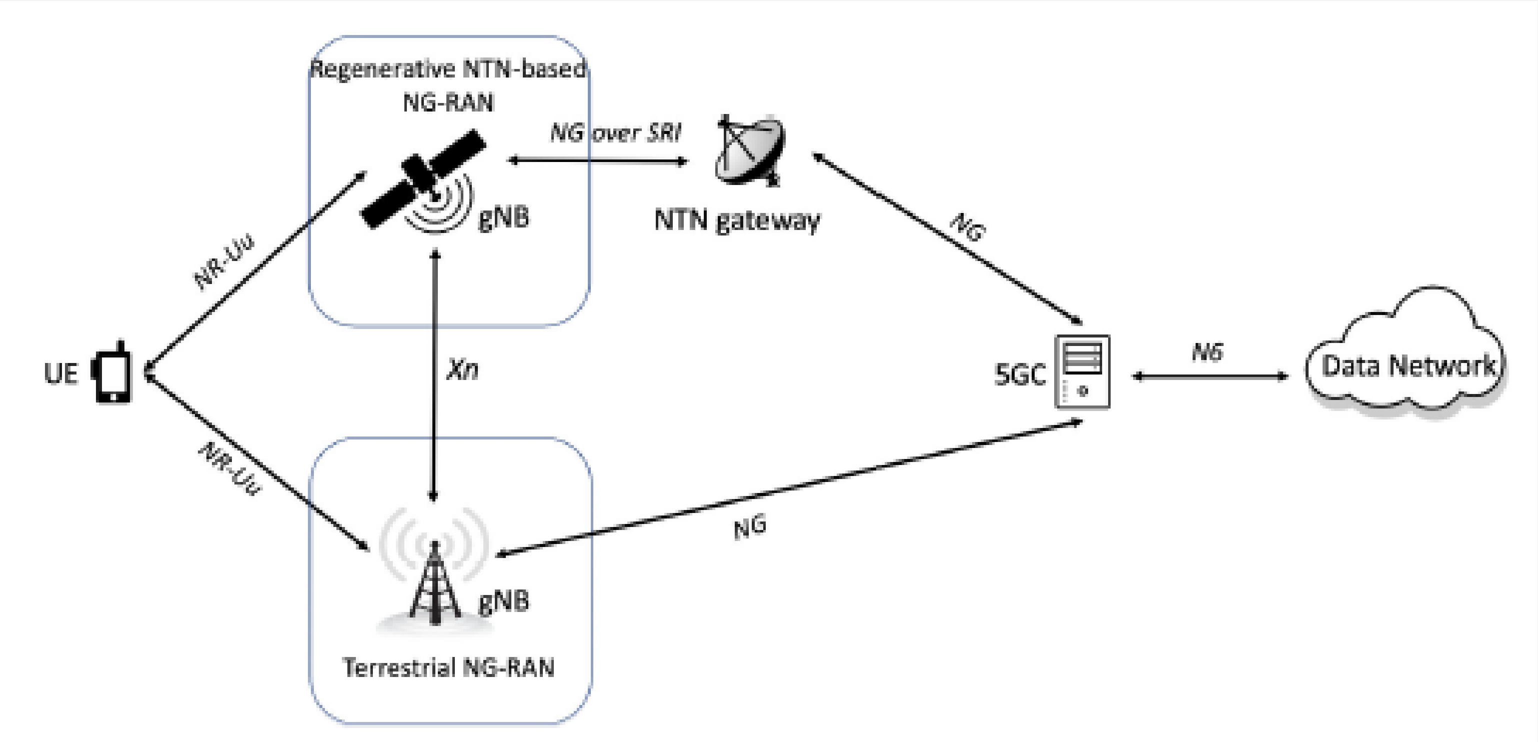
Service Continuity & Multi-Connectivity

- The User terminal is connected simultaneously
  - to the 5GC via two transparent NTN-based NG-RAN



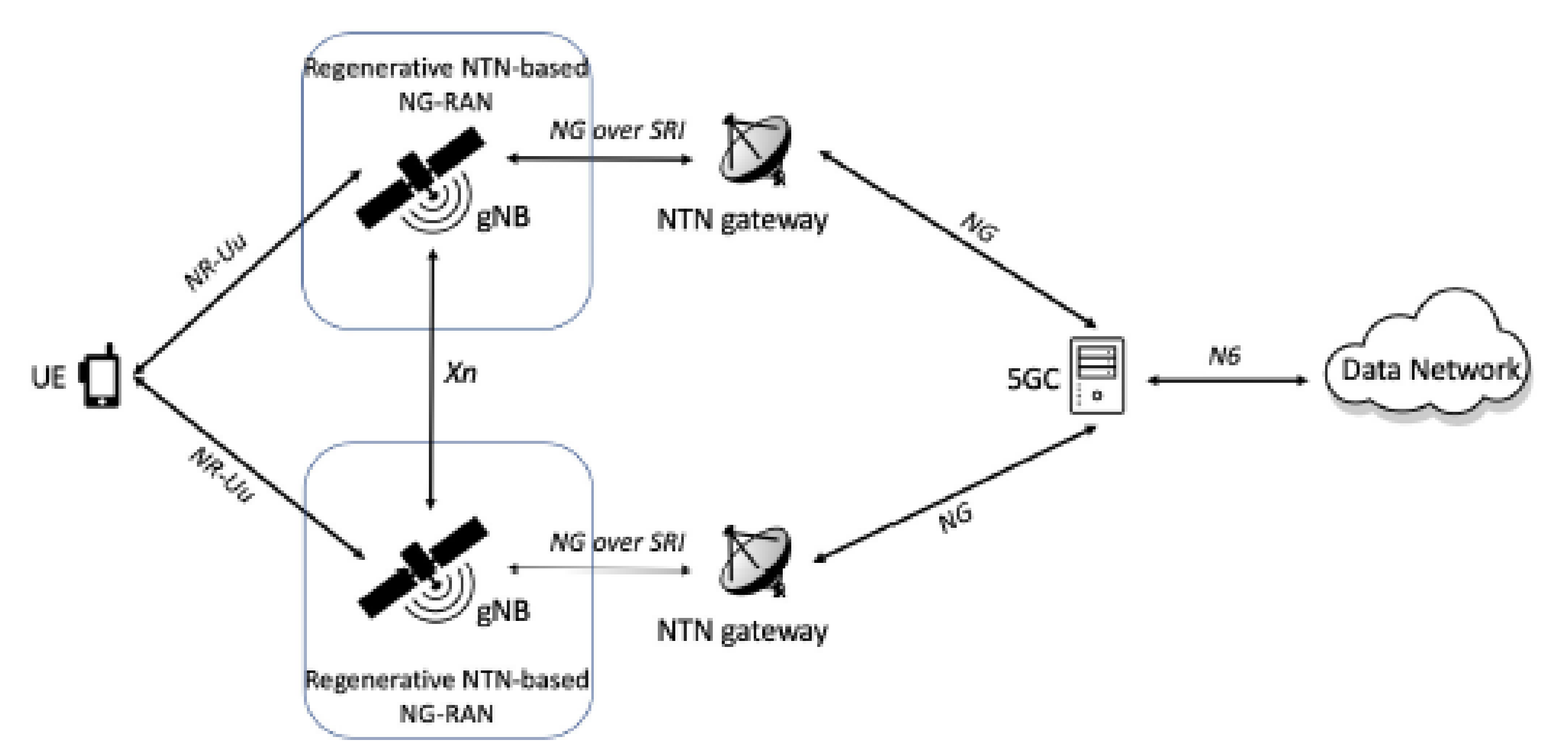
Service Continuity & Multi-Connectivity

- The User terminal is connected simultaneously
  - to the 5GC via both a regenerative NTN-based NG-RAN and terrestrial NG-RAN



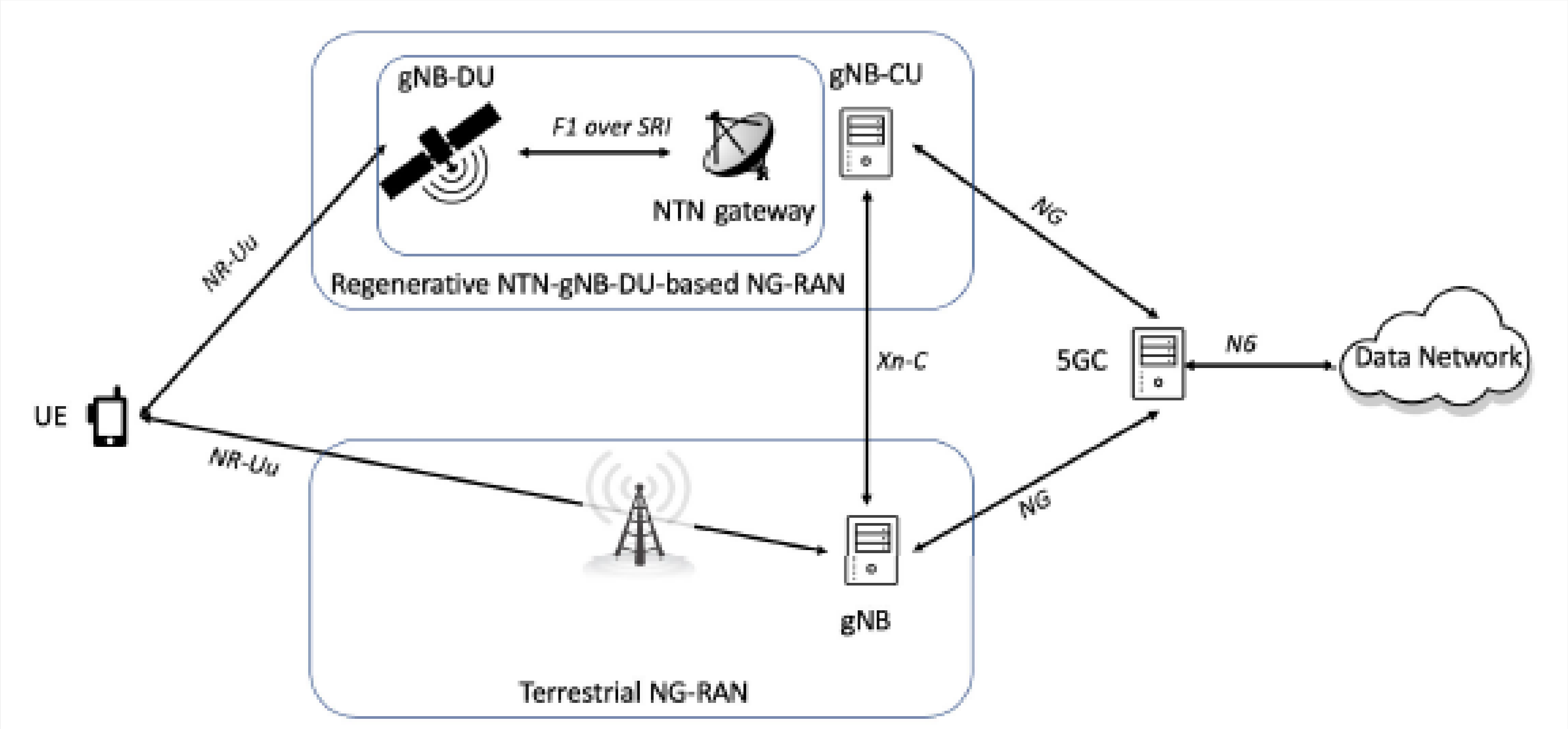
Service Continuity & Multi-Connectivity

- The User terminal is connected simultaneously
  - to the 5GC via two regenerative NTN-based NG-RAN



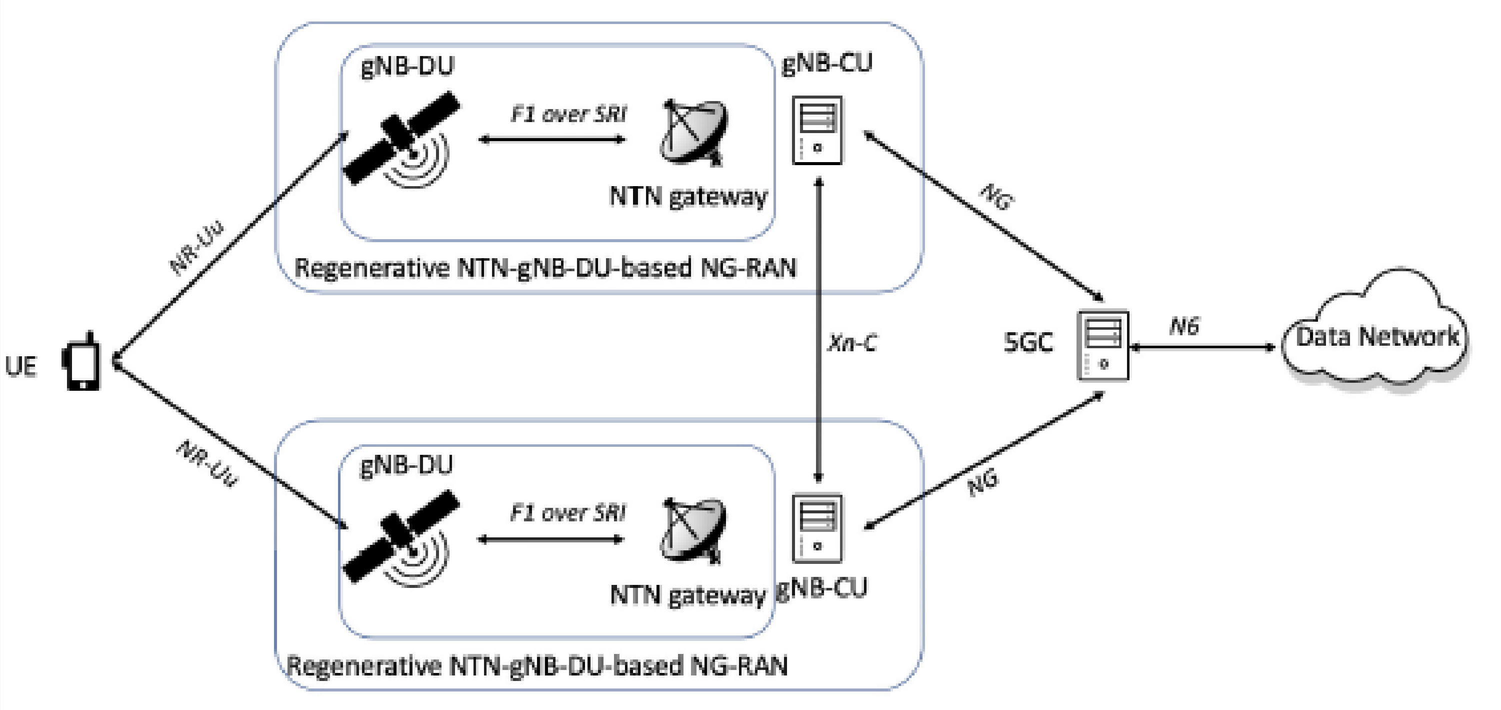
Service Continuity & Multi-Connectivity

- The User terminal is connected simultaneously
  - to the 5GC via both a regenerative NTN-gNB-DU based NG-RAN and terrestrial NG-RAN



Service Continuity & Multi-Connectivity

- The User terminal is connected simultaneously
  - to the 5GC via two regenerative NTN-gNB-DU based NG-RAN



MULTI CONNECTIVITY

- 3GPP requires service continuity
  - between terrestrial and NTN, which can be achieved by
    - mobility scenarios such as inter-RAT handover or by dual connectivity scenarios
    - Running Xn connections between an NTN gNB and a terrestrial gNB is
- With NR-NTN multi connectivity, two scenarios are likely:
  i. Multi connectivity between one NR-NTN access link and one terrestrial-based NR access link
  ii. Multi connectivity between two NR-NTN access links

Multi-Connectivity between NR-NTN and 5G NR

- In this context
  - multi connectivity is defined as the capability for the 5GC to establish
    - for a given UE, simultaneous multiple sessions, each of these sessions taking the benefits of the characteristics (such as coverage, broadcast capability) of different access networks (terrestrial and satellite), in the forward and/or return direction

Technical challenges

- Issues due to
  a. long propagation delays
  b. large Doppler effects
  c. moving cells in NTN

NTN Technical Challenges

- Technical challenges
  - that might have an impact on the NR PHY and MAC layers are
    - propagation delays
    - large Doppler shifts and Doppler rate

NTN Technical Challenges

- The large distance between the terrestrial UE and the satellite
  - has an impact on the link budget or high path attenuation
    - In addition, this huge distance is responsible for the large time delay or RTT, which also varies depending on time and the elevation angle
      - A paradigm change compared to terrestrial networks, where we use the term "base station" to indicate its stationary nature,
        - a LEO satellite travels at a certain velocity which causes a frequency carrier deviation, Doppler shift.
          - And lastly, the ionospheric radio wave propagation is responsible for a rotation of the polarization of the waveform, known in academic circles as Faraday rotation

NTN Technical Challenges

- Round-trip time and differential time delay
  - One of the most challenging NTN characteristics which inhibits the
    - introduction of low-latency communications in NR-NTN is of course
      - the round-trip time (RTT) or long latency due to the large distance between the terrestrial UE and the satellite
        - RTT depends on the elevation angle and is time-variant
          - Typical one-way latency values range from 30ms to 40ms in LEO and up to 544ms in GEO constellations

RTT

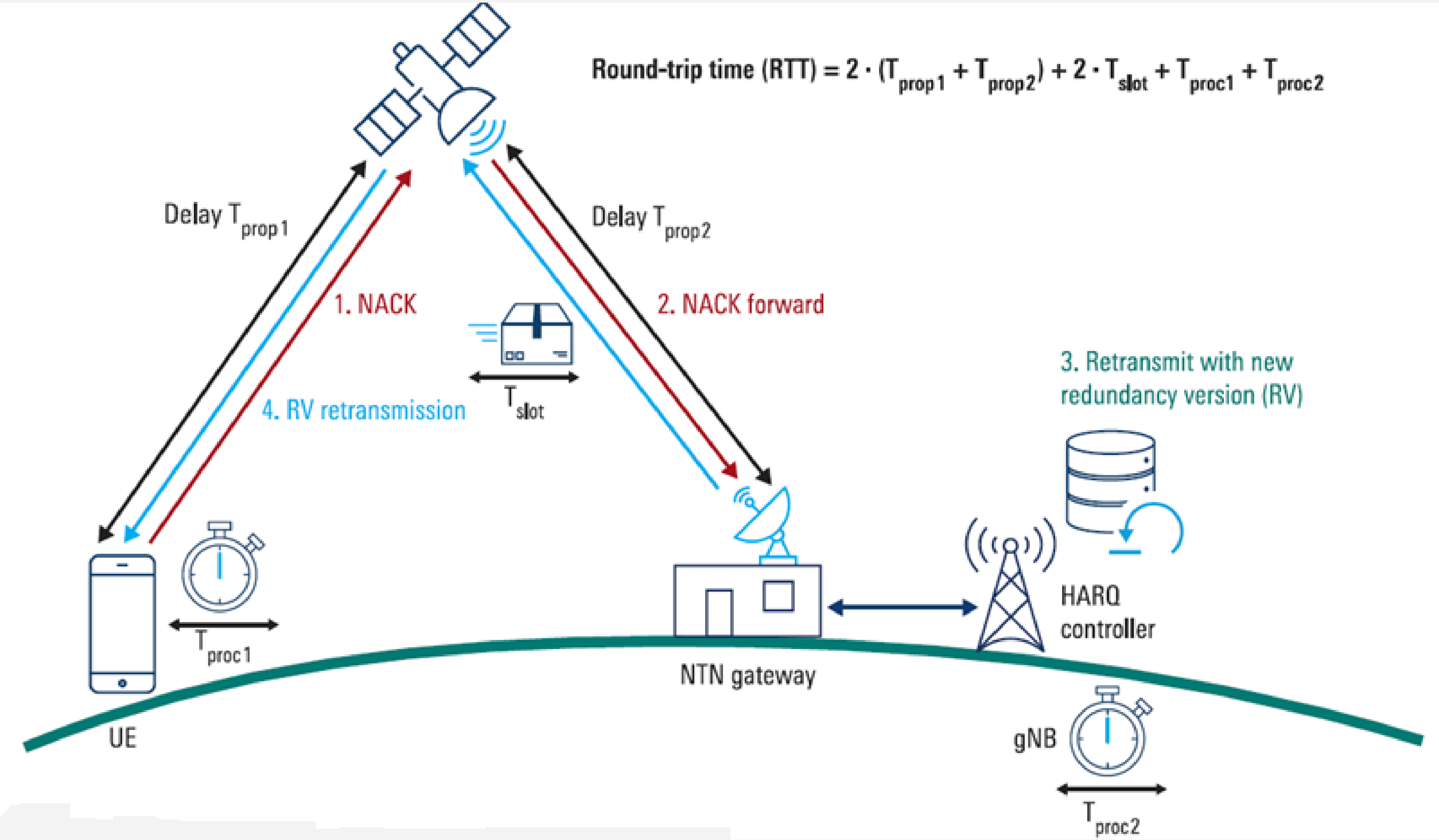
- The round-trip time is influenced by various parameters
  - The assumption is
    - the reception of a PDU at the UE side with negative CRC check
      - Thus, the RTT starts with the NACK message transmission and terminates with the reception of the PDU retransmission
  - propagation delay
    - occurring between UE and satellite and satellite to ground gateway
  - Some system parameters and processing times
    - also need to be considered
      - In the overall RTT
        - the time T_{slot} is the system time corresponding to the duration of a PDU or NACK message and corresponds to subframe or slot durations in 5G

RTT

- Processing time
  - The overall processing time $T_{proc\ 1} + T_{proc\ 2}$ includes
    - the UE and network reaction times
      - The UE reaction time is the time between a CRC failure detection and the transmission of the NACK message.
        - The gNB reaction time is the time between reception of this NACK and transmission of the retransmission PDU

RTT

- NTN: PATH DELAY, ROUND-TRIP TIME RTT



NTN Technical Challenges

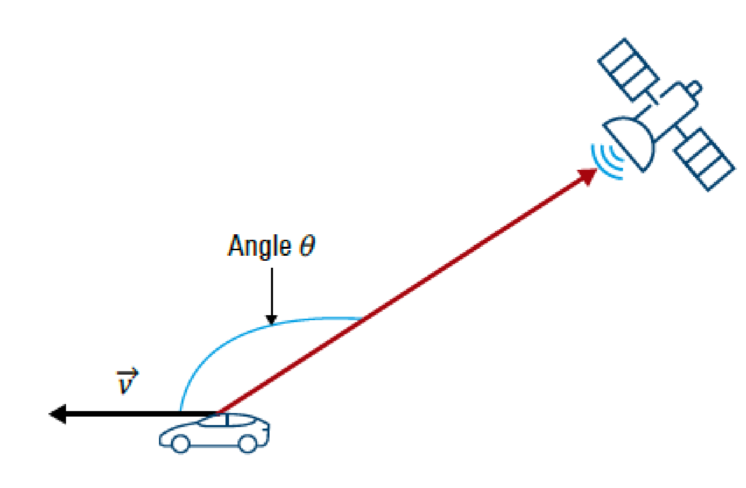
- Carrier frequency shift (Doppler frequency shift)
  - One of the most serious challenges in realizing NTN connections with good end user QoE is
    - the carrier frequency deviation, also known as **Doppler shift**
      - The paradigm change of a moving base station or satellite in combination with
        - an optional moving UE causes a time-variant Doppler shift across the connection time
          - This time-variant Doppler shift is also described as the Doppler variation rate or simply **Doppler rate**

NTN Technical Challenges

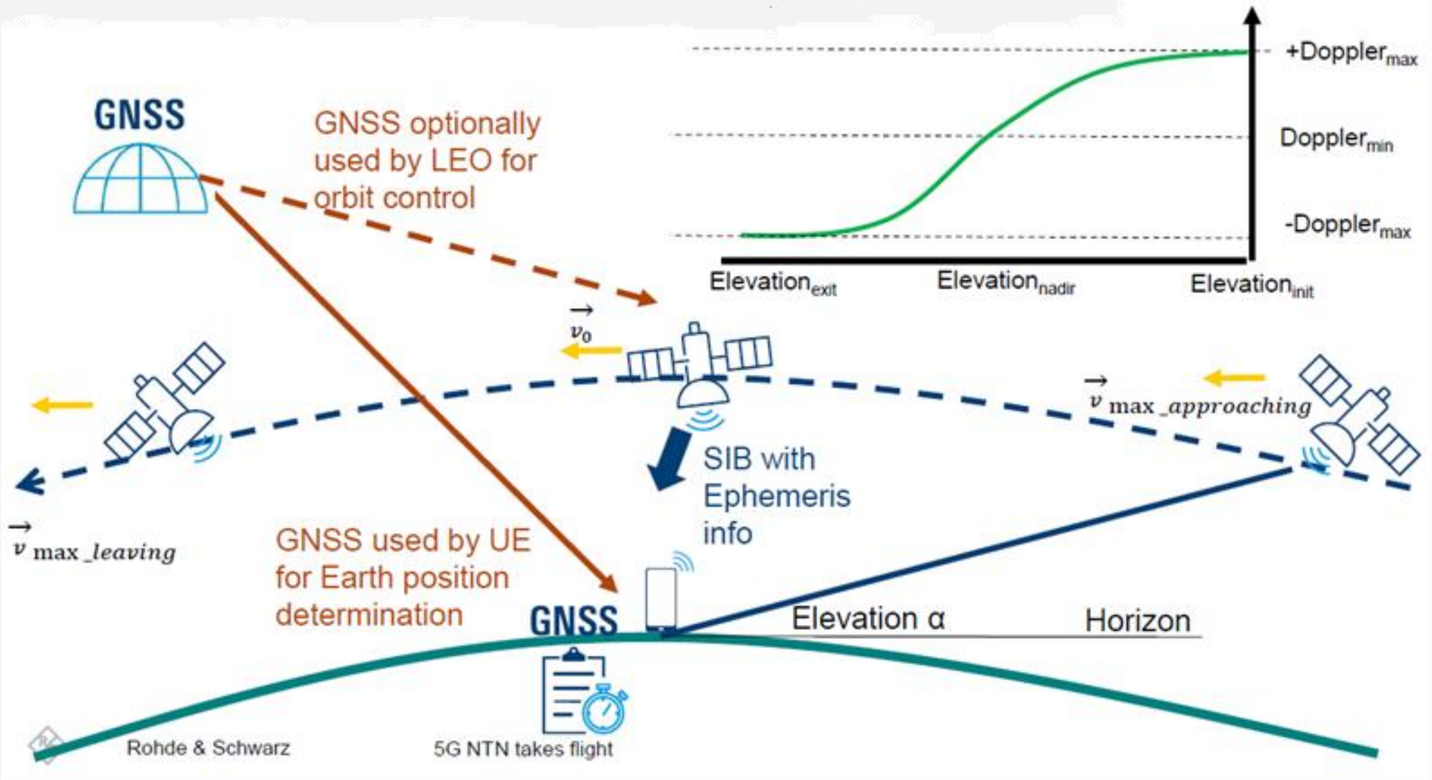
- Carrier frequency shift (Doppler frequency shift)
  - **Doppler rate**
    - In NTN, we do not only have to consider the Doppler shift, which represents the carrier frequency deviation occurring at one instance of time
      - but also the Doppler rate or the Doppler shift variation, because during the connection time, the Doppler shift may vary
    - In a LEO constellation, the elevation angle α changes over the connection period, and the relative velocity of the satellite changes with respect to the UE (see fig)
      - the UE detects the LEO satellite when it rises above the minimum elevation angle at the connection ingress and it loses the satellite once it falls below the minimum elevation angle at the egress of the connection
        - the relative velocity of the satellite with respect to the UE changes from approaching to leaving, causing a Doppler rate

NTN Technical Challenges

- Doppler shift rate in a LEO scenario



$$\Delta f = f_c \cdot v \cdot cos(\theta) / c$$



NTN Technical Challenges

- Doppler shift parameters expected at three different carrier frequencies

Frequency	2 GHz	20 GHz	30 GHz
Doppler shift	±48 kHz	±480 kHz	±720 kHz
Doppler variation	544 Hz/s	5.4 kHz/s	8.1 kHz/s

- This could deeply
  - impact the frequency synchronization of the resources used to transmit through the NR air interface

NTN Technical Challenges

- As a methodology
  - to compensate the Doppler shift in UL
    - the UE estimates its location e.g. based on GNSS positioning, and the UE receives system information (SIB) together with satellite ephemeris information
      - As a result,
        - the UE will then adapt its uplink carrier frequency to adjust the carrier frequency at the satellite receiver in such a way that the Doppler shift impact in uplink direction can be disregarded

NTN Technical Challenges

- Enhancements on
  i. Uplink frequency and time synchronization
  ii. features involving downlink–uplink timing relationships
  iii. Hybrid Automatic Repeat Request (HARQ),